# Forecasting the impact of systematic uncertainties on the dispersion mseasure redshift relation of fast radio bursts

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I hereby declare that this thesis was formulated by myself and that no sources or tools other than those cited were used.

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## Abstract

Fast Radio Bursts (FRBs) are pulses of electromagnetic waves of extragalactic origin. As an FRB propagates through the Universe, its signal gets broadened by the free electrons in the intergalactic medium, leading to a disprsed signal. This dispersion measure (DM) is easily observable via the delayed arrival time which is directly proportional to the column denisty of free electrons, making the DM signal an excellent probe of baryons in the Universe. It is possible to model the DM signal by modelling the cosmological electron number density on large scales; nonetheless, the DM signal receives contribution from the galaxy that hosts the FRB source in addition to a contribution from the Milky Way (MW) galaxy, which poses as systematic uncertainties in a cosmological analysis. Furthermore, redshift measurements and the telescope's sensitivity also represent measurement uncertainty. Failing to account for these effects can lead to biases in the inferred cosmological parameters.

In this thesis, I explore how measurement and systematic uncertainties affect a cosmological analysis and suggest strategies to mitigate their impact for a survey with 10000 detected FRBs. I use the halo model and the baryonification correction model to generate lognormal maps of electron density in bins of redshift in which the mock FRBs are placed. After that, I contaminate the catalog with one of the uncertainties and study its impact of this systematic on the DM-redshift relation. To make the simulation pipeline scalable, I use GPUs to calculate the covariance matrices of the mock data.

I find that ignoring redshift uncertainty leads to more than $1\sigma$ and up to $3.5\sigma$ deviation from the fiducial DM-redshift relation. In addition, ignoring host contrubution leads to $1\sigma$ and up to $4.6\sigma$ deviation. Ignoring the MW contribution leads to up to $1\sigma$ deviation, however, by choosing the appropriate galactic cut, the bias can be reduced without adding a term to the covariance that accounts for the additional MW contribution. For FRBs with unlocalised hosts, I find that unlocalised FRBs at high redshifts lead to severe deviation as compared to unlocalised low redshifts for the same number of unlocalised sources. Finally, I find that it is possible to mitigate all these systematic uncertainties by modelling their impact on the level of the covariance and the FRB catalog.

In total, the efficient simulation pipeline presented in this thesis can be used for simulation-based inference where the systematic uncertainties can be easily put in the simulation to quantify the bias in estimated parameters.

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CHAPTER 1

# Introduction

The questions on how the universe came to be, why the universe is as it is, and what the universe is heading towards fall upon cosmology to answer. Cosmology is the study of the universe on large scales in an attempt to answer the latter questions cite maybe Peebles. This pursuit started as data-starved research cite and transformed into a data-rich endeavour as a result of a plethora of imaging surveys such as the LSST Vera Rubin observatory, Euclid, and Nancy Roman cite. The qualitative questions, with which this paragraph started, translate in the statistical sense to be what is the cosmological model that best describes the universe across all length scales and energy scales.

The first attempt to describe the universe as a whole came through the lens of the general theory of relativity where Robertson and Walker cite built on the work of Friedmann and Lemaître to describe the geometry of the universe through what is now known as the Friedmann-Lemaître-Robertson-Walker (FLRW) metric. We discuss this metric further in 2.2. This metric was largely motivated by the observations of Edwin Hubble who saw that galaxies further away from the Earth are moving faster than galaxies closer to Earth Hubble, 1929. Figure shows the plot from Hubble's work in 1929.

Further observations of the magnitude of distant galaxies have shown that the universe is expanding cite. The general theory of relativity attributes this expansion to the energy density content of the universe. Therefore, the question of what is driving the expansion of the universe and what are the components that contribute to the energy budget of the universe represent a crucial question in cosmology for which multiple cosmological models have been suggested (Di Valentino et al., 2025). In fact, the cosmological model has been motivated by both cosmological and astrophysical observations. Some of these observations are the measurement of galaxy rotation curves by Vera Rubin cite that hinted at an additional matter component that does not interact with light and the discovery of the cosmic microwave background (CMB) radiation that hinted at a hot and homogeneous early universe.cite. Furthermore, early observations of clusters of galaxies have shown that the spatial distribution of galaxies deviates from randomness. This was shown by Peebles cite. By calculating the two point correlation function, there was clearly a signal in the way that galaxies are distributed which points out to an underlying mechanism in the formation of these galaxies (Totsuji and Kihara, 1969). The development of observational techniques and instrumentation allowed the launch of the Center for Astrophysics Redshift Survey (Davis, Huchra et al., 1982) which showed that galaxy positions are correlated on large scales and that there is a structure in the universe (Davis and Peebles, 1983).

In the early 1990s, cosmology experienced a paradigm shift with the launch of the Cosmic Background Explorer (COBE) satellite (Boggess et al., 1992). The COBE mission provided the first detection of

primordial CMB anisotropies and further confirmed that the universe is flat (Smoot, 1999). In the context of the FLRW metric, the flatness of the universe requires a total sum of the relative density parameters to be 1 (within error bounds). Therefore, in addition to the matter component (dark and baryons), there should be a third component that can account for the missing energy density. Using supernova data from the Hubble Space Telescope (Bahcall and Spitzer, 1982), A. G. Riess, Filippenko et al. (1998) found strong evidence for an expanding universe with negative acceleration and supported the findings of the COBE mission for the existence of an additional density component different from matter which is driving the expansion of the universe. Hereafter, the term 'dark energy' was coined to refer to this component (Turner, 2001). With this, the ΛCDM model was born, and the efforts to obtain precise estimates of the cosmological parameters intensified (Bennett et al., 2003; Pope, 2004).

While early experiments relied on CMB, galaxy number counts, or supernova to infer cosmological parameters, gravitational lensing was in development as a tool to constrain the cosmological parameters. Gravitational lensing is the distortion of light geodesics by gravity when light travels from a distant source through the inhomogeneous universe towards an observer on the Earth (P. Schneider, Ehlers and Falco, 1992). On the level of a galaxy, gravitational lensing effects manifest themselves as a distortion of the shape of the galaxy and a change in the angular position with respect to a particular observer. Kaiser and Squires (1993) showed that these distortions could be used to obtain a map of the integrated mass density along the line of sight. The use of galaxy shape distortions for cosmology was a direct result of the Kaiser-Squires inversion formula where Seitz and P. Schneider (1996) realised that the expectation value of galaxy shape is an unbiased estimate of the shear in the weak field limit. In principle, it is possible to relate the power spectrum of galaxy shapes with the matter power spectrum, offering a concrete method of probing cosmology by studying the correlations between galaxy shapes (Bartelmann and P. Schneider, 2001). This new probe of the large scale structure is called cosmic shear and was used by Maoli et al. (2001) to obtain constraints on the cosmological parameters. These constraints were not as tight as the ones obtained by COBE, however, it paved the road towards using cosmic shear to constrain the cosmological model. The advantage of cosmic shear as an integrated quantity along the line of sight means that it is sensitive to structure growth in the late universe. Therefore, it represents a very good tool to constrain the equation of state for dark energy (Hu and Tegmark, 1999; Jain and Taylor, 2003).

After that, multiple galaxy surveys used cosmic shear for cosmological parameter inference, and numerous statistical tools were developed to deal with the increasing data size (P. Schneider, Van Waerbeke et al., 1998; P. Schneider, Eifler and Krause, 2010). Likewise, with increasing data size, the noise level in a cosmological signal increases and the need to mitigate the impact of the systematic and measurement uncertainties is imperative. In this work, we focus on tackling one of the most crucial systematics, namely baryonic feedback on the large scale structure.

Baryonic feedback are the set of baryonic physics processes that impact the distribution of matter (Rudd, Zentner and Kravtsov, 2008). These processes include, but are not limited to, supernova feedback, active galactic nuclei feedback, and star formation. By analysing hydrodynamical simulations, Daalen et al. (2011) realised that baryonic feedback affects the matter distribution in the universe at different scales, meaning that their impact should be modelled and included in a theory of structure formation. However, the theoretical formalism of structure growth, and therefore the matter power spectrum, assumes that structure growth is only driven by gravitational effects, ignoring baryonic feedback (Dodelson and Schmidt, 2026). Modelling baryonic feedback analytically is a cumbersome task, especially since the physics of galaxy formation and evolution is highly uncertain where different suites of hydrodynamical simulations have different feedback models, and each one of these models yields a totally different

description of the matter power spectrum suppression (Daalen et al., 2011). The mismodelling of the matter power spectrum tunnels down to the modelled cosmic shear signal and yields biased parameter estimates in a cosmological analysis.

On a conquest for precision cosmology, multiple statistical and modelling methods attempted to mitigate the impact of baryonic feedback on the large scale structure. For example, Krause et al. (2021) imposed scale cuts on the measured shear correlation function and only considered larger scales where baryonic feedback is minimal. However, the cosmic shear signal derives most of its sensitivity from non-linear scales, i.e. small scales (Piccirilli et al., 2025). Therefore, imposing a scale cut leads to loss of information. Other statistical techniques involve the use of principal component analysis to compress the parameter space of baryonic feedback parameters from hydrodynamical simulations and marginalising over these parameters in a cosmological analysis (Eifler et al., 2015). Another statistical method involves introducing an additive term to the covariance matrix called the theoretical error covariance (Moreira et al., 2021; Maraio, Hall and Taylor, 2025) which describes the variation of different baryonic feedback scenarios among all available hydrodynamical simulations. Analyses involving all the aforementioned results yield unbiased estimates of the cosmological parameters, however, it comes at a cost of introducing additional scatter due to the existing nuisance parameters. Finally, A. Schneider and Teyssier (2015) introduced the baryonification correction model, a model that parametrizes the displacement of dark matter particles in the existence of baryonic feedback. This model and its developments (A. Schneider, Kovač et al., 2025) were calibrated on hydrodynamical simulations and were able to emulate baryonic feedback in other hydrodynamical simulations that they were not calibrated on. Having this model, it is possible to obtain a prediction for any probe of baryons in the universe. In particular, this model was used to constrain baryonic feedback parameters with X-ray and the kinematic Sunyaev-Zel'dovich effect (Kovač et al., 2025) since it is possible to obtain a prediction of gas power spectra from which a simulated signal of the corresponding baryon tracer could be obtained. Though these probes are promising, they rely on some assumptions with respect to the structure of the galaxy clusters such as having hydrodynamic equilibrium and require many convolutions from a prediction of the gas power spectrum up to a predicted signal. Luckily, fast radio bursts provide a direct probe of the electron density in the universe through the information that their signal encodes.

Fast radio bursts (FRBs) are pulses of electromagnetic waves of unknown extragalactic origin. First discovered in 2007 (Lorimer et al., 2007), the most probable canditate for the sources of FRBs is the magnetar which is a star remenant that underwent a supernova. The signal of FRBs gets dispersed while travelling through the universe, meaning that the wave packets of lower frequency will arrive later than wave packets of higher frequency. This difference in arrival time is directly proportional to the integrated electron density along the line of sight where the dispersion measure signal could be easily simulated with minimal assumptions about the relationship of the dispersion measure and the electron number density in the universe. The grand benefit of the dispersion measure signal lies in its constraining power of baryonic feedback parameters since it is a direct tracer of the baryon distribution in the universe. Even though the source of these bursts is unknown (Petroff, J. W. Hessels and Lorimer, 2019), it is still possible to exploit the dispersion measure signal in a cosmological context, given that FRBs are hosted within galaxies whose redshift can be identified.

Simulated likelihood analyses showed that the cross-correlation of FRBs with other probes of structure yields better constraints on some cosmological parameters. In particular, calibrating baryonic feedback with FRBs and weak lensing led to one order of magnitude better constraints on the baryonic feedback and 50% improvements on constraints of the sum of neutrino masses $\sum m_\nu$ and the baryonic fraction in the universe $\Omega_b$ (Reischke, Neumann et al., 2023). Furthermore, combining FRBs with galaxy number

counts in a 3x2-point analysis provided better constraints on baryonic feedback parameters as compared to galaxy number counts (Sharma, Krause et al., 2026). In practice, we can aim to use FRBs alone as a probe to impose tight constraints on the baryonic feedback parameters and use these constraints as priors in a cosmological analysis involving probes of structure, allowing the information content of LSS data to constrain the cosmological parameters and delegating the task of constraining baryonic feedback to FRBs.

The promising potential of FRBs in constraining baryonic feedback could be hindered by multiple modelling and measurement uncertainties. First, FRBs are observed using radio telescopes with only limited follow-up optical observations that are necessary to calculate the redshifts of galaxies and in turn quantify the associated mean DM at that redshift. Second, measurements of these redshifts are highly uncertain especially for sources at high redshifts. Third, the measured dispersion measure signal receives three main contributions: a contribution from the Milky Way, a contribution from the host galaxy, and a contribution from the circumgalactic medium which is the quantity of cosmological interest (Reischke and Hagstotz, 2025). The models for both the Milky Way and the host galaxy contribution are both uncertain, leading to biases in the parameter constraints if not dealt with correctly. Fourth, the theoretical formalism of fast radio bursts assumes that the radio signal travels in a homogeneous universe, ignoring any gravitational lensing effect due to large scale structure. There are contradicting results on the impact of gravitational lensing; Alonso (2021a) showed that first order contributions from general relativistic effects to the DM angular power spectrum are subdominant compared to the contribution from the electron column density, meanwhile, Takahashi (2024) showed that the second order contribution due to lensing could lead to a 1% deviation in the luminosity distance, resulting in an underestimation of the luminosity distance.

Therefore, the purpose of this thesis is to model, forecast, and explore mitigation strategies of systematic uncertainties and describe their importance with varying sample size. With upcoming radio surveys, a plethora of data will be available such that an accurate modelling of the dispersion measure signal is crucial to extract meaningful information from the signal (Luke Hoffmann et al., 2024). I scale up previous work by Konar et al. (2025) by using simulation-based inference (SBI) to infer the impact of systematic uncertainties on the baryonic parameter constraints using fast radio bursts only. Using SBI in this context has multiple benefits. First, it facilitates integrating different uncertainty effects on the level of the dispersion measure signal either on the level of the dispersion measure or the level of the redshift. Second, it removes the necessity to approximate the likelihood of the data given the parameters, unlike traditional Bayesian analyses. Results from ray tracing in hydrodynamical simulations (Konietzka et al., 2025) suggest that a Gaussian or lognormal approximation to the probability distribution function breaks down for large numbers of FRBs, and an analytic model of the dispersion measure is needed to simulate the PDF of FRBs (Torkamani et al., 2026). More details on SBI and the method used in this thesis can be found in 3.5

This thesis is structured as follows: I review the theoretical background in 2 including the general theory of relativity (2.1), the FLRW metric (2.2), the cosmological model (2.3), weak gravitational lensing (2.7), baryonic feedback on the large scale structure (2.8), the basics of fast radio bursts (2.9), and the systematics uncertainties in modelling the DM signal (2.10). I describe the methodology in 3 including modelling the electron power spectrum (3.1), simulating the dispersion measure signal (3.2), describing the systematic uncertainties (3.3), and explaining simulation-based inference (3.5). In 4, I describe the results. Finally, I discuss the results in 5

CHAPTER 2

# Theoretical Background

This chapter is mainly based on 2 textbooks and 1 lecture notes. The first book is *Modern Cosmology by* Dodelson and Schmidt (2026) and the second is *Spacetime and Geometry: An Introduction to General Relativity* by Carroll (2024). The first lecture notes comes from the *Advanced Topics in Cosmology* course given in the Summer semester 2024 by Prof. Dr. Cristiano Poriciani and Jun-Prof. Andrina Nicola. In this chapter, I will introduce the principles of cosmology, cosmic shear, and fast radio bursts (FRBs).

## 2.1 The general theory of relativity

The general theory of relativity introduces a geometric interpretation of gravity, resting on a set of geometric assumptions and principles evolved from the older classical Newtonian theory of gravity and relativity. The first of these is the **weak equivalence principle** which states that **' the inertial mass and gravitational mass of any object are equal'**. Einstein reformulated this statement to fit the language of the special theory of relativity by stating that **'In small enough regions of spacetime, the laws of physics reduce to those of special relativity; it is impossible to detect the existence of a gravitational field by means of local experiments.'** The consequences of this principle are that freely falling frame of reference cannot recognize that freely falling test particles in the vicinity of the frame are being accelerated (or under the influence of gravity). The consequences of this principle invites the formulation of a geometric theory of gravity where gravity manifests itself as curvature.

To have a mathematical theory where observation could be interpreted, Einstein used differentiable manifolds to describe the four dimensional spacetime. Differentiable manifolds allow a definition of physical quantities such as position and momentum along with a continuous derivative of these quantities on the manifolds. Having a differentiable quantity allows also to introduce conservation laws which are a direct results of symmetries in a particular problem (Noether, 1983). Therefore, to describe the motion of particles in a curved spacetime, the metric tensor is defined on a non-Riemannian manifold. The metric tensor is a solution for the Einstein field equations which are second order differential equations on $\mathbb{R}^{4\times4}$.

The metric tensor $g_{\nu\mu}$ (in Einstein index notation) is a $4 \times 4$ symmetric tensor that describes the physical length of an infinitesimal line given a particular system of coordinates such that

$$\mathrm{d}s^2 = g_{\nu\mu}\mathrm{d}x^\nu \mathrm{d}x^\mu, \tag{2.1}$$

where $\{\mathrm{d}x^\mu\}_{0,1,2,3}$ is a basis for the contangent bundle of the manifold. In classical terms, $\{\mathrm{d}x^\mu\}_{0,1,2,3}$ is the differential along a coordinate axis $\{x^\mu\}_{0,1,2,3}$ For a generalized space time, the path of a free particle is described by the geodesic equation. If the coordinates of a particle in spacetime $x^\mu$ is parameterized with an affine parameter $\lambda$, then the coordinates $x^\mu(\lambda)$ is constrained by an ordinary differential equation, namely the geodesic equation given as:

$$\frac{\mathrm{d}^2 x^\mu}{\mathrm{d}\lambda^2} + \Gamma^\mu_{\rho\sigma}\frac{\mathrm{d}x^\rho}{\mathrm{d}\lambda}\frac{\mathrm{d}x^\sigma}{\mathrm{d}\lambda} = 0, \tag{2.2}$$

where $\Gamma^\mu_{\rho\sigma}$ are the Christoffel symbols given in terms the metric tensor such that

$$\Gamma^\sigma_{\mu\nu} = \frac{1}{2}g^{\sigma\rho}\left(\partial_\mu g_{\nu\rho} + \partial_\nu g_{\rho\mu} - \partial_\rho g_{\mu\nu}\right). \tag{2.3}$$

In the Newtonian limit, the gravitational potential $\Phi$ is related to the mass density $\rho$ in the following equation

$$\nabla^2\Phi = 4\pi G\rho, \tag{2.4}$$

with $G$ being the gravitaional constant. To formulate a theory which reduces to Eq. 2.4, we define the energy-momentum tensor $T_{\mu\nu}$ which is a a $4\times 4$ symmetric tensor that describes the energy density and shear in the four-dimensional spacetime. Since the gravitational field manifest itself as curvature, the analogy to Eq. 2.4 motivate a proportionality relationship between the second derivative of the metric tensor $g_{\mu\nu}$ and the stress-energy tensor $T_{\mu\nu}$

$$\left[\nabla^2 g\right]_{\mu\nu} \propto T_{\mu\nu}. \tag{2.5}$$

A geometric quantity that is composed of the second derivatives of the metric tensor is the Ricci tensor $R_{\mu\nu}$ expressed in terms of the Christoffel symbols

$$R_{\mu\nu} = \partial_\lambda\Gamma^\lambda_{\nu\mu} - \partial_\nu\Gamma^\lambda_{\lambda\mu} + \Gamma^\lambda_{\lambda\rho}\Gamma^\rho_{\nu\mu} - \Gamma^\lambda_{\nu\rho}\Gamma^\rho_{\lambda\mu}. \tag{2.6}$$

Meanwhile, the Ricci scale is defined as the contraction of the Ricci tensor

$$R = R^\mu_\mu = g^{\mu\nu}R_{\mu\nu}. \tag{2.7}$$

Therefore, a possible relationship between the stress-energy tensor and the Ricci tensor is

$$R_{\mu\nu} = \kappa T_{\mu\nu}. \tag{2.8}$$

To find $\kappa$, the conservation of energy is invoked as

$$\nabla^\mu T_{\mu\nu} = 0, \tag{2.9}$$

where $\nabla^\mu$ is the covariant derivative. Following Eq. 2.8 and using Eq. 2.6 and Eq. 2.7, the covariant

derivative of the Ricci tensor is

$$\nabla^{\mu} R_{\mu\nu} = \frac{1}{2} \nabla^{\mu} R g_{\mu\nu}. \tag{2.10}$$

Therefore, we can define another tensor $G_{\mu\nu}$ with a vanishing covariant derivative

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu}. \tag{2.11}$$

$G_{\mu\nu}$ is called the Einstein tensor, and the Einstein's field equation regardless

$$G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}, \tag{2.12}$$

where I used $\kappa = 8\pi G/c^4$ which can be obtained by reducing Eq. 2.12 to the Newtonian limit in Eq. 2.4. Einstein modified Eq. 2.12 to account for a vacuum solution by adding a term to represent the vacuum energy $\Lambda g_{\mu\nu}$ where $\Lambda$ is called the cosmological constant, and the final equation becomes

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G T_{\mu\nu}. \tag{2.13}$$

Finally, the 4-velocity of a particle in general relativity is defined as

$$U^{\mu} = \frac{\mathrm{d}x^{\mu}}{\mathrm{d}\lambda}. \tag{2.14}$$

Therefore, the 4-velocity would follow the normalization

$$g_{\mu\nu} U^{\mu} U^{\nu} = -1, \tag{2.15}$$

and the 4-momentum for a particle of mass $m$ is defined as

$$P^{\mu} = m U^{\mu}, \tag{2.16}$$

and the normalization of momentum is

$$g_{\mu\nu} P^{\mu} P^{\nu} = -m^2. \tag{2.17}$$

Notice that either of the previous normalization is invariant under coordinate transformation. We define the observed energy of a particle of 4-momentum $P^{\mu}$ with respect to an observer with 4-velocity $U^{\mu}$ such that it reduces to the special relativistic limit if the particle has a vanishing spatial velocity and the observer

$$E_{obs} = -g_{\mu\nu} U^{\mu} P^{\nu}. \tag{2.18}$$

The Einstein theory of general relativity can explain cosmological observations such as cosmological redshift and the expansion of the universe where the expansion is driven by the energy content in the universe. Though the FLRW metric (explained in 2.2) is phenomenologically motivated, the variation of the scale factor with conformal time $a$ is described by the Einstein's field equation.

## 2.2 The Friedmann-Robertson-Lemaître-Walker metric

To describe the entire universe using one metric tensor requires educated simplifications and assumptions. The first guiding principle is the Copernican principle (Bondi and Roxburgh, 2010) which states that **'we do not occupy a special place in space'**. The Copernican principle motivates the cosmological principle which states that 'the universe is homogeneous and isotropic'. Early galaxy surveys have shown that the universe is only homogeneous on large scales, therefore, the full cosmological principle is stated as **'the universe is homogeneous and symmetric on large scales'**. An additional assumption that simplifies that narrows down the possible formula for a metric is Weyl's postulate which states that **'the world lines of the galaxies form a 3-sheaf of non-intersecting geodesics orthogonal to layers of spacelike hypersurfaces'**. Weyl's postulate, coupled with the cosmological principle, allows a better definition of where homogeneity and isotropy apply, namely that the universe is 'spatially homogeneous and isotropic'. Therefore, a maximally symmetric metric that follows both principles takes the form

$$\mathrm{d}s^2 = -c^2\mathrm{d}t^2 + g_{ij}\mathrm{d}x^i\mathrm{d}x^j, \tag{2.19}$$

where $t$ is cosmic time. A two-dimensional schematic diagram of this metric is shown in 2.1.

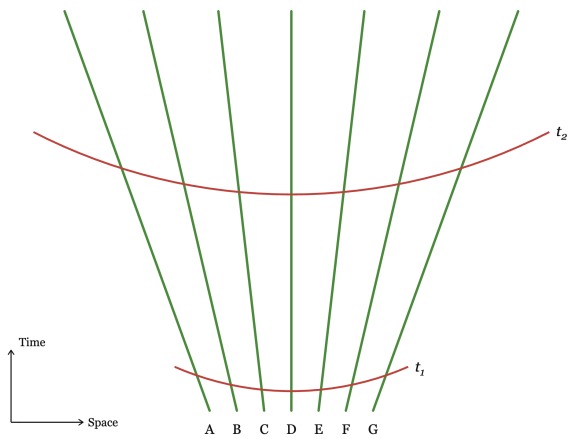



Figure 2.1: A schematic representation of spacetime that follows Weyl's postulate. The green lines labelled with letters are world lines, the path of freely falling objects in the spacetime (or geodesics). The redlines are spacelike hypersurfaces labelled with the same cosmic time $t_i$. These hypersuraces are homogeneous and isotropic.

Using the tools of differential geometry, a maximally symmetric metric that also accounts for a homogeneous expansion of the universe is

$$\mathrm{d}s^2 = -c^2\mathrm{d}t^2 + a^2(t)\mathrm{d}s_3^2, \tag{2.20}$$

where $a(t)$ is the scale factor that couples the spatial part with time and the spatial part, $c$ is the speed of light, $\mathrm{d}s_3^2$ represent a volume element in a maximally symmetric space that takes the following form in hyperspherical coordinates

$$\mathrm{d}s_3^2 = \mathrm{d}\chi^2 + S_k(\chi)^2\left(\mathrm{d}\theta^2 + \sin^2\theta\,\mathrm{d}\phi^2\right), \tag{2.21}$$

the function $S_k(\chi)$ takes one of three forms by virtue of the number $k$

$$S_k(\chi) = \begin{cases} \frac{1}{\sqrt{k}}\sin(\sqrt{k}\chi) & \text{if } k > 0, \text{for a closed universe} \\ \chi & \text{if } k = 0, \text{for a flat universe} \\ \frac{1}{\sqrt{|k|}}\sinh(\sqrt{|k|}\chi) & \text{if } k < 0, \text{for an open universe} \end{cases}. \tag{2.22}$$

The case when $k = 0$ represents a flat universe, where the spatial part reduces to the usual spherical coordinates. The flat FLRW metric reads

$$\mathrm{d}s^2 = -c^2\mathrm{d}t^2 + a^2(t)\left[\mathrm{d}\chi^2 + \chi^2\left(\mathrm{d}\theta^2 + \sin^2\theta\,\mathrm{d}\phi^2\right)\right]. \tag{2.23}$$

The coordinate $\chi$ could be understood by considering two space like points in spacetime, one with $\chi = 0$ and another with $\chi = \chi_c$ leading To

$$s = a(t)\chi_c, \tag{2.24}$$

which means that the physical distance changes with time for a given $\chi_c$. We call $\chi_c$ the comoving distance. Taking the derivative with respect to cosmic time in Eq. 2.24, we can retrieve the empirical law that Hubble (1929) discovered by recording the relationship between the distance to galaxy and the speed of the galaxy

$$\frac{\mathrm{d}s}{\mathrm{d}t} = V = \frac{\dot{a}}{a}s, \tag{2.25}$$

with $\dot{a} = \mathrm{d}a/\mathrm{d}t$. We define the Hubble parameter at a cosmological time $t$ such that

$$H(t) = \frac{\dot{a}}{a}. \tag{2.26}$$

Another distance indicator is gravitational redshift, similar to doppler redshift, gravitational redshift is the modulation of light frequency as light travels across the universe. If a fundamental observer at cosmic time $t_0$ emits a burst of light, then the energy of this light particle is given by Eq. 2.18. If another fundamental observer observes this light at a later time $t$, the ratio between the two energies is defined as

$$1 + z = \frac{E_e}{E_o}, \tag{2.27}$$

where $E_e$ and $E_o$ are the energy of light measured by the emitter and the observer respectively, and **$z$ is called the cosmological redshift**. $z$ is related to the wavelength of light in the limit where each of the emitter and the observer are fundamental observers, meaning that their 4-velocity is $U^\mu = (1, 0, 0, 0)$. Therefore, the light energy could be expressed using Planck's law (Planck, 1900), $E = h/\lambda$, with $h = 6.626 \times 10^{-34}$J·s is Planck's constant, and $\lambda$ is the wavelength of light. The final relationship describing the modulation in the observed wavelength of light is

$$1 + z = \frac{a(t_o)}{a(t_e)} = \frac{\lambda_o}{\lambda_e}, \tag{2.28}$$

with $\lambda_e$ and $\lambda_o$ are the light wavelength measured by the emitter and the observer respectively. This relationship could be exploited by quantifying the modulation of the wavelength and its relationship to

the scale factor. In a nutshell, a light emitted at a scale factor $a(t_e)$ travels through an expanding universe until measured by another observer at time $t_o > t_e$ corresponding to $a(t_o) > a(t_e)$, therefore $\lambda_o > \lambda_e$, which explains why distant sources are redshifted. In practice, by measuring the spectrum of light of a distant galaxy and convolving it with a standard spectrum (the expected spectrum of galaxy in the nearby universe), it is possible to obtain the redshift of the distant galaxy. If we choose $a = 1$ in the current epoch, the relationship between redshift and the scale factor reduces to

$$a = \frac{1}{1+z}, \tag{2.29}$$

where $z = 0$ refers to the current epoch. Since light moves along a null geodesic, it is possible to obtain the comoving distance to an emitter given the redshift of the emitter. Starting from Eq. 2.23, and working in a coordinate system with the light travelling along $\theta = 0$ and $\phi = 0$, the null geodesic dictates that

$$c\mathrm{d}t = a(t)\mathrm{d}\chi, \tag{2.30}$$

which could be further arranged using Eq. 2.26 and Eq. 2.29 into

$$\chi = \int_1^{z_e} \frac{c}{H(z)}\mathrm{d}z. \tag{2.31}$$

From the perspective of Eq. 2.29, the cosmological redshift can be used as proxy of time where larger value of redshfit conrreponds to smaller value of the scale factor $a$. Meanwhile from the perspective of Eq. 2.31, the cosmolgoical redshift is proxy for distance where sources of larger redshift correspond to larger distance. In a cosmological galaxy survey, measuring distances from the Earth towards galaxies is not a trivial task, however, the cosmological redshift can be measured by measuring the deviation the electromagnetic spectrum of a galaxy from the expected spectrum of a galaxy at redshift $z = 0$ [1].

Another distnace scale of interest is the Hubble scale. The Hubble scale is the distance at which the recession velocity of a galaxy equals the speed of light. Using Hubble's law, the Hubble scale is given by

$$s_H = \frac{c}{H}. \tag{2.32}$$

The Hubble scale is a distance scale that is used to define the size of the observable universe. It has a significant consequences on the growth of perturbations in the universe, as will be discussed in 2.8.

Throughout this chapter, a question is lurking behind all the kinematics of the universe that the FLRW metric proposes: how does the scale factor $a$ change with cosmic time? In principle, this requires using the Einstein's field equation Eq. 2.13 along with assumptions about the form of the stress energy tensor $T^{\mu\nu}$. Assuming that the energy density in the universe is that of a perfect fluid, solving the Einstein's field equation for $a$ yields

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}\left(\rho + \frac{3p}{c^2}\right) + \frac{\Lambda c^2}{3} \tag{2.33a}$$

$$\frac{\dot{a}^2}{a^2} = \frac{8\pi G}{3}\rho + \frac{\Lambda c^2}{3} - \frac{kc^2}{a^2}, \tag{2.33b}$$

[1] Of course, the process of measuring redshifts is cumbersome in practice, and this statemetn is meant to explain the basic principle behind measring the cosmological redshift.

with $p$ being the isotropic pressure of the cosmic fluid and $\rho$ being the density [2]. These equations are known as the Friedmann equations, describing the evolution of the scale factor $a$ with respect to cosmic time $t$. To solve these equations, a relationship between the pressure p and the energy density is needed, known as the equation of state. The possible components that contribute to the energy density $\rho$ have been a process of trial and error starting de Sitter universe (Sitter, 1917) which is a universe with no matter up to the recently speculated model with dynamical dark energy (Adame et al., 2025). Eqs. 2.33 were derived with the assumption of a homogeneous and isotropic universe on all scales; this is not the case when studying the inhomogeneous universe with large-scale structure. Therefore, the physical quantities in Eqs. 2.33 are only a functions of time (the total density $\rho(t)$ and the isotropic pressure $p(t)$).In 2.3, I briefly present the case of inhomogeneous universe with perturbations

$$\frac{\rho(t,\mathbf{x}) - \bar{\rho}(t)}{\bar{\rho}(t)} << 1, \tag{2.34}$$

where $\bar{\rho}(t)$ is the average density at a cosmic time $t$, and $\rho(t,\mathbf{x})$ is the density at a comoving position $\mathbf{x}$ and cosmic time $t$. In this thesis, we consider the ΛCDM cosmological model and describe its components in 2.3.

## 2.3 The cosmological model

In this section, we consider the ΛCDM cosmological model and explain its components. As noted by Eqs. 2.33, the time evolution of the scale factor $a$ depends on the density and its relation to the pressure, known as the equation of state. If we refer to the current day value of the Hubble parameter as $H_0$ and the current day value of the density $\rho_0$, Eq. 2.33b can be written in the current epoch

$$1 = \frac{8\pi G}{3H_0^2}\rho_0 + \frac{\Lambda c^2}{3H_0^2} - \frac{kc^2}{H_0^2}. \tag{2.35}$$

Notice that the quantitiy $\frac{3H_0^2}{8\pi G}$ has the dimensionality of a density, and we call it the critical density $\rho_{cr}$. It is possible to express the density fractions of the different components in terms of the current day value of the component's density and the critical density.

$$\Omega_0 = \frac{\rho_0}{\rho_{cr}}, \tag{2.36a}$$

$$\Omega_\Lambda = \frac{\Lambda c^2}{3H_0^2}, \tag{2.36b}$$

$$\Omega_k = -\frac{kc^2}{H_0^2}. \tag{2.36c}$$

We list below the constituents of the ΛCDM model of cosmology. It is common to express the density of a constituent as a fraction of the critical density, such that the sum of these fractions equals 1. In

[2] Since energy and mass are equivalent up to the squared of the speed of light, we may refer to the different constituents of the universe as 'contributing to the energy budget of the universe'.

practice, the density fraction of a constituent $X$ in the current epoch $\Omega_X$ is expressed as

$$\Omega_X = \frac{\rho_{0,X}}{\rho_{cr}}, \tag{2.37}$$

with $\rho_{0,X}$ being the density of a constituent $X$ at the current epoch.

Measurements from the cosmic microwave background has shown that we live in a flat universe (Smoot, 1999), meaning that $\Omega_k = 0$ (within error bounds). **For the rest of this work, I assume a flat universe**. The two remaining components are the density fraction $\Omega_0$ and the fraction $\Omega_\Lambda$. The mechanism of how these components were formed into existence will be omitted from this thesis. It is worth noting that the Hubble parameter $H_0$ is conventionally given in term of a dimensionless number $h$ such that

$$H_0 = 100\, h\, \mathrm{km\, s^{-1}\, Mpc^{-1}}, \tag{2.38}$$

and many other quantities are expressed in terms of $h$; for example, the density ratios $\Omega_X$ could be given as $\Omega_X h^2$. Measurements of the CMB with the Planck satellite (Aghanim et al., 2020) reported a result of

$$H_0 = 67.66 \pm 0.42 \mathrm{km\, s^{-1}\, Mpc^{-1}},$$

meanwhile measurements using supernovae from the SH0ES collaboration (A. G. Riess, Yuan et al., 2022) report a value of

$$H_0 = 73.04 \pm 1.04 \mathrm{km\, s^{-1}\, Mpc^{-1}}.$$

The discrepancy of $H_0$ measurements between the early universe with the CMB and late universe measurements with supernovae is called the Hubble tension (Verde, Schöneberg and Gil-Marín, 2024).

The equation of state of each component in the ΛCDM model differs between the early universe with large energy scales and the late universe with very low energy scales. We list below each component and its role in the cosmic evolution of the universe. In general, the equation of state $p_X(\rho_X)$ expresses the relationship between pressure $p_X$ and density $\rho_X$ of a constituent $X$. Motivated by dimensional analysis [3], we can conclude that the pressure and density are related

$$p_X = w \rho_X c^2. \tag{2.39}$$

The relationship between the scale factor $a$ and the density $\rho_X$ could be found by invoking the first law of thermodynamics and solving the following ordinary differential equation,

$$\mathrm{d}(\rho_X c^2 a^3) = -p_X\, \mathrm{d}(a^3). \tag{2.40}$$

In total, the ΛCDM universe contains the following constituents:

- **Radiation:** Radiation is mostly composed of freely propagating relativistic particles: photons and neutrinos. Photons have been propagating in the universe since recombination. Recombination is the epoch when free electrons combine with nuclei to form neutral atoms, after which the photon mean free path becomes effectively infinite and photons decouple from matter; these photons are observed today as the cosmic microwave background (CMB) radiation. CMB anisotropies contain

[3] Dimensional analysis is the method of comparing the units of two physical quantities to conclude the relationship between them up to a proportionality constant.

a wealth of information about the density fractions of the other components of the Universe and its spatial curvature. The temperature of the photon background scales as $T_r \propto a^{-1}$. With equation of state

$$p_r = \frac{1}{3}\rho_r c^2, \tag{2.41}$$

and the radiation energy density scales as $\rho_r \propto a^{-4}$.

$$\rho_r = \rho_{r,0} a^{-4}. \tag{2.42}$$

In addition to photon, neutrinos contribute to the radiation budget in the universe. At early times, neutrinos are in equilibrium with other particles such as electrons and photons, however, as the universe expands and the temperature drops, electrons no longer interact with neutrinos, therefore, neutrinos decouple from the rest of the components in the universe in a process known as freeze-out. The density of neutrinos scales to the scale factor in a similar way to photons, however, the present day temperature of neutrinos is different from that of photons

- **Baryonic matter:** Baryonic matter is composed of the particles that make up the standard model of particle physics. The particles that make up this component interact with light and with itself through long range pressure. In the early universe, baryonic matter is coupled to photons through absorption, emission, and scattering processes. Due to the expansion of the universe, the absorption cross-section approaches zero, therefore, the photons emitted from recombining electrons with nuclei have less energy than needed to ionize a similar atom, leaving the photon free. The process of forming neutral atoms by combining free electrons with nuclei is called recombination. This process continues until all the atoms are neutral, and then, photons propagate freely in the universe, with a very low probability of interacting with Baryons. After an epoch of structure formation (with the universe being transparent), neutral gas clouds in dark matter halos collapse under gravity to form stars and trigger fusion reactions [4]. Photons emitted from stars reionize the medium around stars, and baryons interact again with photons (Draine, 2011). The fraction of baryonic matter in the universe is measured to be $\Omega_b h^2 = 0.0225 \pm 0.0003$. In the early universe, the equation of state of baryonic matter is derived with relativistic consideration, however, after recombination, the short range pressure of baryons is negligible and the equation of state for baryons is that of dark matter on large scales. On small scale, baryonic processes in the late universe are highly non-linear and cumbersome to model which impose a limitation on the modelling of the cosmic shear signal. **This is the exact problem that this thesis is trying to tackle**.

- **Dark matter:** Dark matter is matter that does not interact with light, however, its effect is observed gravitationally. Recent results from different cosmological probes including CMB, galaxy clustering and weak lensing favour a cold dark matter component, meaning that the temperature of dark matter is much less than its mass. As a consequence, dark matter could be modelled as a pressureless fluid on large scale (on small scales, the fluid approximation breaks down). Using this simple picture of cold dark matter (CDM), it is possible to solve the fluid equation of dark matter to linear order and obtain a good approximation of the linear density

[4] A fusion reaction is a nuclear reaction when two low mass atoms combine to form a heavier atom, emitting an amount of energy.

evolution with cosmic time (I show this derivation below). The equation of state of dark matter, therefore, is simply $p = 0$. By solving Eq. 2.40 for dark matter, the density of dark matter depends on the scale factor $a$ such that

$$\rho_c = \rho_{c,0} a^{-3}. \tag{2.43}$$

The present day value of the dark matter density fraction is measured to be $\Omega_{\rm m} h^2 = 0.1431 \pm 0.0025$ by the Planck collaboration (Aghanim et al., 2020). Dark matter is the main driver of matter clustering in the universe since dark matter can form virialized region of space called dark matter halos which accretes baryons and leads to galaxy formation as a consequence (Wechsler and J. L. Tinker, 2018).

- **Dark energy:** dark energy is a term coined by Michael S. Turner (Turner, 2001) to describe the contribution to the energy budget that is not in a form of particles. From Eqs. 2.36, dark energy is the component with density fraction $\Omega_\Lambda$. Early on, this component was considered to be smooth and homogeneous, meaning that its corresponding density $\rho_\Lambda = \Omega_\Lambda \rho_{cr}$ only varies with cosmic time. In its simplest form, the dark energy pressure is related to the dark energy density as follows

$$p_\Lambda = w_\Lambda \rho_\Lambda c^2, \tag{2.44}$$

with the simplest case $w_\Lambda = -1$. The equation of state of dark energy is highly debated with many modification suggested to explain tensions in cosmology such as the Hubble tension (Verde, Schöneberg and Gil-Marín, 2024). In this work, we assume an equation of state of dark energy with $w_\Lambda = -1$. The Planck collaboration (Aghanim et al., 2020) reported that $\Omega_\Lambda = 0.6889 \pm 0.0056$.

Figure 2.2 shows a schematic representation of the components discussed above. In general, the matter density fraction $\Omega_{\rm m}$ is the sum of the baryonic matter density fraction $\Omega_{\rm b}$ and the dark matter density fraction $\Omega_c$. The total energy budget of the universe is the sum of the matter density fraction $\Omega_{\rm m}$, the dark energy density fraction $\Omega_\Lambda$, and the radiation density fraction $\Omega_r$. Since the radiation density fraction is negligible in the current epoch, we have $\Omega_{\rm m} + \Omega_\Lambda \approx 1$. With that, the evolution of the expansion rate of the universe in terms of the scale factor $a$ is given by

$$H^2(a) = H_0^2 \left[ \Omega_r a^{-4} + \Omega_{\rm m} a^{-3} + \Omega_\Lambda \right], \tag{2.45}$$

where the different additive terms dominate the evolution at different epochs. In particular, dark energy is the dominant component in the current epoch. To constrain the equation of state of dark energy, cosmic shear stands as a powerful probe of late structure formation since it is an integrated qunatity that includes signficant contribution from the late evolution of the universe, and therefore sensitive to dark energy. The basic principles of cosmic shear and its relationship to the matter density field are explained in 2.7. This thesis focuses on using FRBs to constain baryonic feedback on the large scale structure and therefore improve the modelling of cosmic shear on small scales, which in turn improves the constraining power of cosmic shear on dark energy.

So far, I have only introduced the universe as a homogeneous entity, ignoring the fact the universe is inhomogeneous on small scales. As a physical system, the initial conditions of the universe are scale-dependent Gaussian perturbations which grow to be highly non-linear (and non-Gaussian) large perturbations (large relative to the mean density of the universe). Since the initial perturbations were generated via a stochastic process, the resultant perturbations are called random fields. In particular, the

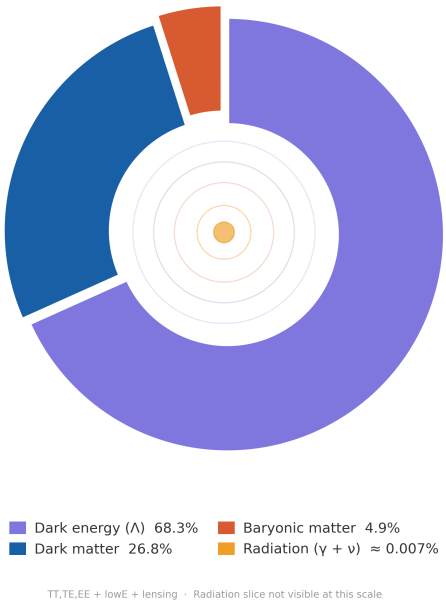



Figure 2.2: A schematic representation of density fractions of the different components in the ΛCDM model. The density fractions are measured by the Planck collaboration (Aghanim et al., 2020) using measurements of the CMB anisotropies.

matter density field $\rho_{\mathrm{m}}(\mathbf{x}, t)$ is the random field that describes the matter density at a comoving position $\mathbf{x}$ and cosmic time $t$. For simplicity, the matter perturbations are often expressed in terms of the matter density contrast $\delta_{\mathrm{m}}$ such that

$$\delta_{\mathrm{m}}(\mathbf{x}, t) = \frac{\rho_{\mathrm{m}}(\mathbf{x}, t) - \bar{\rho}_{\mathrm{m}}(t)}{\bar{\rho}_{\mathrm{m}}(t)}, \tag{2.46}$$

where $\bar{\rho}_{\mathrm{m}}(t)$ is the average matter density at a cosmic time $t$. In addition, similar definitions apply to any other density field such as the baryonic density contrast or the electron density contrast, which will be important in the study of FRBs. By construction, any density contrast field is a random field with a mean of zero i.e. $\langle \delta_{\mathrm{m}} \rangle = 0$. Higher order statistics of a random field also encode information about the random field. This thesis only uses the two-point statistics of the matter density contrast and its Fourier space counterpart, namely the power spectrum. We discuss these statistics below. Since dark matter is the dominant component of matter in the universe, it is the component responsible for the observed structure in the universe. In addition, the pressure induced by baryonic processes is negligible on large scales, meaning that the evolution of matter density contrast on large scales is well approximated by the evolution of dark matter density contrast. Therefore, the evolution of matter density contrast $\delta_{\mathrm{m}}$ is well approximated by the evolution of dark matter density contrast $\delta_c$. I present below the Einstein-Boltzmann equation that describes the phase-space evolution of a particle species in the universe.

## 2.4 The Boltzmann equation & the linear perturbation theory

This section is partly based on a review on the effective field theory of large scale structure (EFTofLSS) by Tobias Baldauf (Baldauf, 2020). The Boltzmann equation describes the evolution of the phase-space distribution function of a particle species in the universe. This function is labelled as $f(\mathbf{x}, \mathbf{p}, t)$, where $\mathbf{x}$ is the position, $\mathbf{p}$ is the momentum, and $t$ is the cosmic time. Different observables could be obtained from the distribution function $f$ including the density $\rho(\mathbf{x}, t)$ and the velocity field $\mathbf{v}(\mathbf{x}, t)$. The Boltzmann equation for a collisionless and pressureless dark matter

$$\begin{aligned} \frac{\mathrm{d}f}{\mathrm{d}t} &= \frac{\partial f}{\partial t} + \frac{\mathrm{d}\mathbf{x}}{\mathrm{d}t} \cdot \frac{\partial f}{\partial \mathbf{x}} + \frac{\mathrm{d}\mathbf{p}}{\mathrm{d}t} \frac{\partial f}{\partial \mathbf{p}} \\ &= \frac{\partial f}{\partial t} + \frac{\mathbf{p}}{ma} \cdot \frac{\partial f}{\partial \mathbf{x}} - am\nabla\phi \cdot \frac{\partial f}{\partial \mathbf{p}} = 0, \end{aligned} \tag{2.47}$$

with $m$ being the mass of the dark matter particle. If dark matter behaves as a fluid, the density and the bulk velocity of the fluid could be obtained by taking the first two moments of the distribution function $f$ such that

$$\rho_{\mathrm{m}}(\mathbf{x}, t) = \frac{m}{a^3} \int \mathrm{d}^3 p \, f(\mathbf{x}, \mathbf{p}, t), \tag{2.48}$$

$$v_i(\mathbf{x}, t) = \int \mathrm{d}^3 p \, \frac{p_i}{am} f(\mathbf{x}, \mathbf{p}, t) \Big/ \int \mathrm{d}^3 p \, f(\mathbf{x}, \mathbf{p}, t), \tag{2.49}$$

$$\sigma_{ij}(\mathbf{x}, t) = \int \mathrm{d}^3 p \, \frac{p_i}{am} \frac{p_j}{am} f(\mathbf{x}, \mathbf{p}, t) \Big/ \int \mathrm{d}^3 p \, f(\mathbf{x}, \mathbf{p}, t) - v_i(\mathbf{x}) v_j(\mathbf{x}), \tag{2.50}$$

with $\sigma_{ij}(\mathbf{x}, t)$ representing the velocity dispersion tensor. The distribution function $f$ has higher order moments, however, the anisotropic stress tensor is second order in momentum, therefore, it can be postulated that it is negligible at linear order and at early times where the velocity dispersion is small[5]. In this thesis, I am only interested in the evolution of the matter density contrast. Using Eq. 2.47, the evolution of the matter density contrast and the bulk velocity are governed by the continuity and the Euler equations. The equations up to linear order are as follows

$$\frac{\partial \delta_{\mathrm{m}}}{\partial \tau} + \frac{1}{a} \nabla \cdot \mathbf{v} = 0, \tag{2.51a}$$

$$\frac{\partial \mathbf{v}}{\partial \tau} + \frac{1}{a} (\mathbf{v} \cdot \nabla) \mathbf{v} = -\frac{1}{a} \nabla \phi, \tag{2.51b}$$

with $\tau$ being the conformal time defined as $\mathrm{d}\tau = \mathrm{d}t/a$. Eqs. 2.51 show that at a linear order the matter density contrast is coupled to the velocity field, and the velocity field is coupled to the gravitational field $\phi$. The growth of structure is, therefore, sourced by gravity.

To solve the equations in real space is a cumbersome task, therefore, we take the following steps in order to obtain the evolution of the matter density contrast.

- Apply the Fourier transform to Eqs. 2.51 to obtain the equations in Fourier space. The operator $\nabla$

[5] The usual practice in standard perturbation theory is to truncate the Boltzmann hierarchy even for higher order perturbations

reduces to a multiplication by $i\mathbf{k}$, with $\mathbf{k}$ being the Fourier space wavevector.

- The velocity field is a vector field, therefore, it can be decomposed into a curl-free component and a divergence-free component. The curl-free component is called the velocity divergence $\theta$ and is defined as $\theta = \nabla \cdot \mathbf{v}$, while the divergence-free component is called the vorticity $\omega$ and is defined as $\omega = \nabla \times \mathbf{v}$. At linear order, the vorticity decays with time, therefore, it can be neglected at linear order.

- The peculiar gravitational potential $\phi$ can be related to the matter density contrast by the relationship known as the Poisson equation such that

$$\nabla^2 \phi = \frac{3}{2}\Omega_{\rm m} H_0^2 \frac{\delta_{\rm m}}{a}. \tag{2.52}$$

With these steps, the equation that governs the eveolution of the matter denstiy field at a linear level is a second order ordinary differential equation with respect to the conformal time $\tau$ such that

$$\frac{{\rm d}^2\delta_{\rm m}}{{\rm d}\tau^2} + \mathcal{H}\frac{{\rm d}\delta_{\rm m}}{{\rm d}\tau} - \frac{3}{2}\Omega_{\rm m} H_0^2 \frac{\delta_{\rm m}}{a} = 0, \tag{2.53}$$

with $\mathcal{H} = aH$ being the conformal Hubble parameter. A solution for the matter density contrast $\delta$[6] at linear order in Fourier space as follows

$$\delta(\mathbf{k}, t) = D_+(\tau)\delta_{+,0}(\mathbf{k}), \tag{2.54}$$

where $\delta_{+,0}$ is the initial conditions of the matter density contrast, and $D_+(\tau)$ is the growth factor whose exact expression depends on the cosmological model considered. For an Einstein-de Sitter universe, which is a universe with $\Omega_{\rm m} = 1$ and $\Omega_\Lambda = 0$, the growth factor is simply $D_+(\tau) = a(\tau)$. The importance of the result 2.54 lies in two observations: the first is that the growth of each Fourier mode of the matter density contrast is independent from other modes (modes of different length scales are not coupled), and the second is that the time evolution of the matter density contrast is separable from its spatial dependence.

The solution to Eqs. 2.51 is given assuming that the universe is way into matter dominated epoch, where the impact of radiation is negligible. In the early universe, the radiation component is dominant and initiates pressure that only acts on sacles smaller that the horizon scale, defined in Eq. 2.32, since pressure is a causal process that can only propagate at a speed less than the speed of light. Therefore, $\delta_{+,0}(\mathbf{k})$ should encode all the physics of the early universe including the impact of radiation on the matter density contrast and the initial conditions. In practice, comparing the evolved density field with observaiton will not give a conclusive answer on whether the theory presented here is correct or not, since the current universe is one realisaiton of the stochastic process that generated the initial conditions, therefore, estimating the statistical properties of the observed matter density field and comparing it with the theoretical prediction is the way to test the theory. In the next section, I present the power spectrum as a statistic that encodes the statistical properties of the matter density field.

At early times, the amplitude of the matter density contrast is small, therefore, the linear solution in Eq. 2.54 is a good approximation to the evolution of the matter density contrast. As time goes on, the

[6] $\delta$ refers to the matter density contrast $\delta_{\rm m}$ hereafter, and the subsript is omitted for ease of writing.

amplitude of the matter density contrast grows and the linear approximation breaks down. Furthermore, for regions in the universe where the average density within this specfic region is large compared to the average density of the universe, dark matter paricles collapse under the influence of gravity and virialised regions of space form. These virialised regions are called dark matter halos. The appendix include an explanation of the toy model for the collapse. add the number of the appendix. Baryons fall in the deep potential well of the dark matter halo and accretes to form galaxies, meaning that galaxies traces the position of dark matter halos, and in turn dark matter.

## 2.5 The 2-point correlation function and the power spectrum

Since $\delta_0(\mathbf{k})$ is a Gaussian random field, its statistical properties are completely determined by its mean and variance. Starting from $\delta_0(\mathbf{k})$, the matter density contrast in real space reads

$$\delta(\mathbf{x}) = \frac{1}{2\pi}\int \delta(\mathbf{k})e^{i\mathbf{k}.\mathbf{x}}\mathrm{d}\mathbf{k}. \tag{2.55}$$

The mean of $\delta_0(\mathbf{k})$ is zero by construction, therefore, the variance is the only quantity that encodes the statistical properties of $\delta_0(\mathbf{k})$. The variance of $\delta_0(\mathbf{k})$ is called the power spectrum and is defined as

$$\langle\delta_0(\mathbf{k})\delta_0^*(\mathbf{k}')\rangle = (2\pi)^3 P(k)\delta_D(\mathbf{k}-\mathbf{k}'), \tag{2.56}$$

where $\delta_D$ is the Dirac delta function. Notice that the power spectrum is only a function of the magnitude of the Fourier space wavevector $k = |\mathbf{k}|$ due to the statistical isotropy of the universe. Fig. 2.3 shows the matter power spectrum obtain using the CLASS Boltzmann solver (Lesgourgues, 2011) with varying fraction of baryonic matter $\Omega_\mathrm{b}$ and varying Hubble parameter $H_0$ at redshift $z = 0$.

The real space counterpart of the power spectrum is called the 2-point correlation function $\xi(r)$, and is defined assume

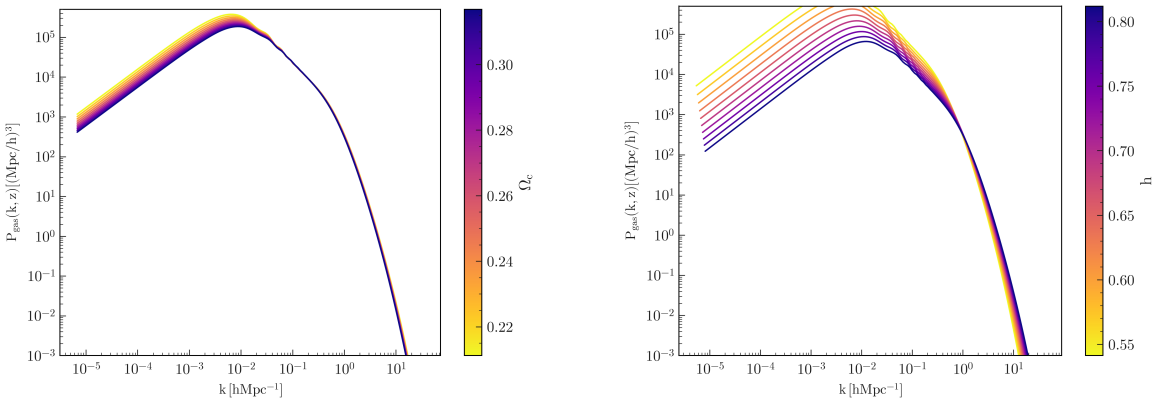
$$\xi(r) = \langle\delta(\mathbf{x})\delta(\mathbf{x}-\mathbf{r})\rangle \tag{2.57}$$

where $\xi$ is a function of the comoving lenth scale $r$ and is independent of direction. In practice, $\xi$ represent the excess probability that two points separated with $r$ have denisty contrast of $\delta(\mathbf{x})$ and $\delta(\mathbf{x}-\mathbf{r})$ respectively.

### 2.5.1 Dark matter tracers

Though the matter power spectrum is a powerful statistic that encodes the statistical properties of the matter density field, it is not directly observable. In order to relate the matter field to obervables, it is important to understand the physical mechanisms by which the observables are related to the matter field. For galaxies, galaxy formation theories suggest that baryonic matter falls into the potential well of a dark matter halo and accretes to form a galaxy. Fig. 2.4 illustrastes a snapshot of hydrodynamical simualtion with a dark matter halo hosting a galaxy.

The detailed mechanism of how this happens is highly non-linear and complicated. In addition, running comprehensive galaxy formation simulations is computationally expernsive in order to cover the large parameter space of galaxy formation physics (Somerville and Davé, 2015). Therefore, it is common to use a phenomenological model to relate the galaxy field to the matter field. The simplest model is the linear bias model which assumes that the galaxy number density contrast $\delta_g$ is linearly




(a) (b)

Figure 2.3: A plot of the matter power spectrum at redshift $z = 0$ with varying fraction of cold dark matter density $\Omega_c$ and varying Hubble parameter $h$. The power spectrum is obtained using the CLASS Boltzmann solver (Lesgourgues, 2011). Notice the the power specturm peaks at a scale equivalent to the Hubble scale at the matter-radiation equality. Furthermore, the wiggles in the power spectrum are due to the sound waves that popagates in the baryon-phton fluid due the contrasting force of the fluids's pressure and gravity.

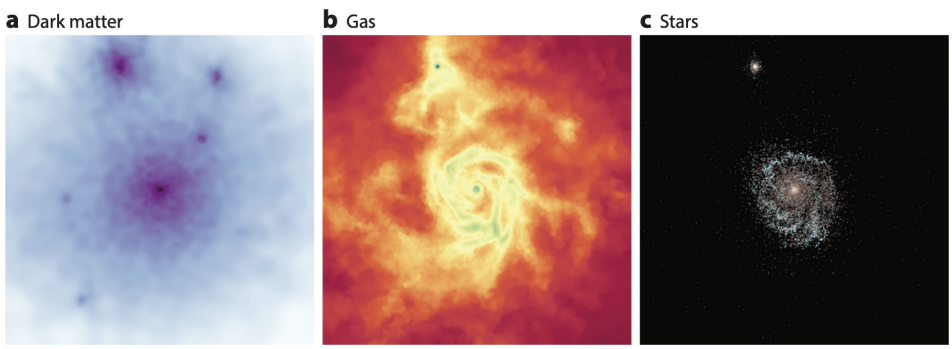



Figure 2.4: A 2D plot from the IllustrisTNG simulation (Pillepich et al., 2018), adopted from Crain and Voort (2023). It is a snapshot of hydrodynamical simulation showing how a 'danser than average' region of dark matter hosts a galaxy (stars make up the 'skeleton' of a galaxy).

related to the matter density contrast $\delta$ such that

$$\delta_g(\mathbf{x}, t) = b(t)\delta(\mathbf{x}, t), \tag{2.58}$$

with $b$ being the linear bias parameter. The linear bias model is a good approximation on large scales, however, on small scales, the relationship between the galaxy field and the matter field is highly non-linear and the linear bias model breaks down (Desjacques, Jeong and Schmidt, 2018). Notice that the galaxy number density contrast is only dependent on the local matter density field and does not depend on the environment surrounding the galaxy. This approximation breaks down on small scales where th environment of a galaxy can have a significant impact of the galaxy formation process. Galaxies are called 'biased tracers of dark matter' since they trace the underlying matter field. For any tracer u of dark

matter, I define the pwoer spectrum of the density constrast of this tracer $\delta_{\mathrm{u}}$ in a similar fashion to Eq. 2.56

$$\langle \delta_{\mathrm{u}}(\mathbf{k})\delta_{\mathrm{u}}^{*}(\mathbf{k}')\rangle = (2\pi)^3 P_{uu}(k)\delta_D(\mathbf{k}-\mathbf{k}'). \tag{2.59}$$

It is possible to use the same formalism of galaxy bias to relate the matter density contrast with other tracers of dark matter such as dark matter halos. In practice, the bias of dark matter halos is related to the underlzing matter field via the following relationship

$$\delta_h(\mathbf{x},t) = b_h(t)\delta(\mathbf{x},t), \tag{2.60}$$

with $b_h$ being the dark matter halo bias parameter. Owing to Eq. 2.60, the halo power psectrum can be expressed in terms of the matter power spectrum defined in Eq. 2.56 such that

$$P_{hh}(k,t) = b_h^2(t)P_{mm}(k,t). \tag{2.61}$$

For dark matter halos, it is possible to go bayond a linear bias model by fitting an empirical model of halo bias to the results of an N-body gravitional simulation. In this thesis, I use the fitting formula of J. L. Tinker et al. (2010) to model the halo bias. In this recipe, the halo bias is a function of the halo mass and its redshift $b_h(M,z)$. As a consequence, the halo power spectrum is related to the matter power spectrum with the relationship

$$P_{\mathrm{hh}}(M_1,M_2,k) = b(M_1)b(M_2)P_{\mathrm{mm}}^{\mathrm{lin}}(k)[1+\beta^{\mathrm{nl}}(M_1,M_2,k)]. \tag{2.62}$$

In the next section, I present the halo model, a non-pertubative approximation to obtain the non-linear power spectrum of matter or a tracer of matter. In particular, this thesis is concerned with the electron power spectrum in order to model the cosmological contribution of the dispersion measure of fast radio bursts.

### 2.5.2 The angular power specrtum

The definition of the power spectrum in Eq. 2.56 assumes that the observed field is sampled in three-dimensional space, therefore, the density field in real space can be written in using the Fourier space exponent as in EQ. 2.55 which represent a full orthogonal basis. WHen observing cosmological fields on the sky, a different othogonal basis is needed to decompose the cosmological field since the values measured live on the unit spehre and not on a flat geometry. For this purpose, the a cosmological field $f$ in the direction $\boldsymbol{\theta}$ can be written in terms of the spherical harmonic functions $Y_{\boldsymbol{\ell}}(\boldsymbol{\theta})$, where $\boldsymbol{\ell}$ is the Fourier space counterpart of $\boldsymbol{\theta}$. Therefore, if $f_{\boldsymbol{\ell}}$ is the spherical harmonic coeficients and $\boldsymbol{\ell} = (\ell, m)$[7] for a moment $m$ and a multipole $\ell$, the spherical harmonic transform and its inverse are given as

$$f(\boldsymbol{\theta}) = \sum_{\ell m} Y_{\ell m}(\boldsymbol{\theta})\, f_{\ell m} \quad \leftrightarrow \quad f_{\ell m} = \int \mathrm{d}\boldsymbol{\theta}\, f(\boldsymbol{\theta})\, Y_{\ell m}^{*}(\boldsymbol{\theta}). \tag{2.63}$$

The angular power spectrum $C_\ell$ is then defined in terms of the spherical harmonic coefficients

$$\langle \hat{f}_{\ell m}\hat{f}_{\ell' m'}\rangle = \delta_{\ell\ell'}\delta_{mm'}C_\ell, \tag{2.64}$$

[7] more on the spherical harmonics can be found in the appendix

where $\delta$ here is the Kronecker delta. Notice how the power spectrum is not dependent on $m$ by virtue of the cosmological principle.

The angular power specrtrum of a cosmological field can be obtained from the power spectrum of the same cosmological field or an integral of it. I show how to get the angular power spectrum from the electron power spectrum in 2.9.

## 2.6 The halo model & the electron power spectrum

The details in this section are based on a review of the halo model by Asgari, Mead and Heymans (2023). The halo model assumes that matter in the universe are entirely contained in dark matter halos. Therfore, any cosmological field $\delta_{\mathrm{u}}$ at a comoving coordinate $\mathbf{x}$ can be expressed in terms of the halo denisty profile $W_{\mathrm{u},i}$ such that [8]

$$\delta_{\mathrm{u}}(\mathbf{x}) = \sum_i N_i \, W_{\mathrm{u},i}\left(\left|\mathbf{x} - \mathbf{x}_i\right|\right) , \tag{2.65}$$

where the sum runs over all volume elements and $N_i = \{0, 1\}$ determines whether there is a halo centre in that volume element. $W_{\mathrm{u},i}$ could express the halo density profile for any comp onent that traces halos such as electrons. In N-body simulations, the halo profile $W_{\mathrm{u},i}$ of a particular dark matter halo $i$ difffers from one halo to another as a funciton of the halo mass $M$. To express this scatter, the halo-mass distribution function is introduced. The halo mass distrinution function $n(M)$ is defined as the number of dark matter halos with $M_i \in [M, M + \mathrm{d}M]$ is $n(M)\mathrm{d}M$. Averaging over all possible halo-masses, the expectation value of a field becomes

$$\langle \delta_{\mathrm{u}}(\mathbf{x}) \rangle = \left\langle \sum_i N_i \, W_{\mathrm{u}}(M_i, |\mathbf{x} - \mathbf{x}_i|) \right\rangle = \int_0^\infty \mathrm{d}M \, n(M) \int \mathrm{d}^3x' \, W_{\mathrm{u}}(M, |\mathbf{x}-\mathbf{x}'|) = \int_0^\infty \mathrm{d}M \, W_{\mathrm{u}}(M) n(M) , \tag{2.66}$$

where $W_{\mathrm{u}}(M, \mathbf{x})$ can be split into two multiplicative component: $W_u(M)$ which encodes the amplitud eof the halo profile and $U_{\mathrm{u}}(M, \mathbf{x})$ whichc encodes information about the shape of the profile, andd the mass-dependent halo profile can be written as

$$W_{\mathrm{u}}(M, x) = W_{\mathrm{u}}(M) U_{\mathrm{u}}(M, x), \tag{2.67}$$

with $W_{\mathrm{u}}(M)$ appearing in the last equality, and $U_{\mathrm{u}}(M, x)$ is regourously defined as

$$\int \mathrm{d}^3x \, U_{\mathrm{u}}(M, x) = 1 . \tag{2.68}$$

The physical intuition behind the function $W_{\mathrm{u}}(M, x)$ can be understood by going to Fourier space. First, Eq. 2.68 indicates that the Fourier space counterpart of the function $U_{\mathrm{u}}(M, x)$, namely $\hat{U}_{\mathrm{u}}(M, k)$ has the property $\hat{U}_{\mathrm{u}}(M, k \to 0) = 1$ which implies that $\hat{W}_{\mathrm{u}}(M, k \to 0) = \hat{W}_{\mathrm{u}}(M)$. THis result is understood that at large scales, dark matter halos act as point masses (similar to a Dirac-delta function) which translates to a constant in Fourier space.

With this formalism in place, the ingredients of the halo model allows an esitmation for the power spectrum of a species u (in the case of this thesis, the halo model formalism will be applied to obtain a

[8] notice that the time dependency is omitted for ease of writing.

prediction of the electron density in the universe). Using the definition of the pwoer spectrum in Eq. 2.56, the power spectrum of the field $\delta_u$ as described inthe halo model is

$$\langle \hat{\delta}_{\mathrm{u}}(\mathbf{k}) \hat{\delta}_{\mathrm{u}}^{*}(\mathbf{k}') \rangle = \left\langle \sum_{i,j} e^{-i\mathbf{k}\cdot\mathbf{x}_i} \, e^{i\mathbf{k}'\cdot\mathbf{x}_j} \, N_i \, N_j \, \hat{W}_{\mathrm{u},i}(M,k) \, \hat{W}_{\mathrm{u},j}(M,k') \right\rangle . \tag{2.69}$$

The sum can be further divided into two contributions:

- a contribution with $i = j$ which is the power restultant from the correlation of dark matter particles within one dark matter halo given by

$$P_{\mathrm{uu}}^{\mathrm{1h}}(k) = \int_0^\infty \hat{W}_{\mathrm{u}}(M,k) \, \hat{W}_{\mathrm{u}}(M,k) \, n(M) \, \mathrm{d}M \, , \tag{2.70}$$

  and this contribution is termed as **the one-halo term**.

- a contrubution with $i \neq j$ whcih is the power resultant from the correlation of dark matter particles in two different halos given as

$$P_{\mathrm{uu}}^{\mathrm{2h}}(k) = \int_0^\infty \int_0^\infty \mathrm{d}M_1 \, \mathrm{d}M_2 \, P_{\mathrm{hh}}(M_1, M_2, k) \, \hat{W}_{\mathrm{u}}(M_1,k) \, \hat{W}_{\mathrm{u}}(M_2,k) \, n(M_1) \, n(M_2), \tag{2.71}$$

  and this contribution is termed as **the two-halo contribution**. $P_{hh}$ is the halo power spectrum defined in Eq. 2.62.

The non-linear power spetrum wihtin the halo model of a species u is given by

$$P_{uu}(k) = P_{\mathrm{uu}}^{\mathrm{1h}}(k) + P_{\mathrm{uu}}^{\mathrm{2h}}(k). \tag{2.72}$$

Fig. 2.5 shows an example of the non-linear power specrutm with the different contributions from the one-halo and the two-halo terms. Since $P_{\mathrm{uu}}^{\mathrm{1h}}$ describes the **inter-halo** clustering of a species u, it dominates the power spctrum on small scales. Meanwhile,

Naturally, the halo-model is a recipe for obtainting an approximnation of the non-linear pwoer spectrum. By knowing the halo profile of a spcies u, it is possible to obtain the non-linear power spectrum of this species. In this thesis, we use the halo model to obatin a theoretical prediction of the electron power spectrum. Section 3.1 describes the procedure of obtaining the electron power spectrum using the baryonfication correction model, and section 2.8 describes how electron profiles are obtained usinga model for baryonic feedback.

Nonetheless, the halo model has its shortcoming (Mead et al., 2021a). The power spectrum estimation via the halo model agrees with the power specrtum estimation from N-body gravitional simualtions on very large and very small scales. Meanwhile, it fails on intermediate scales, the scales where the transition from the two- to the one-halo term. However, for the use case specific to this thesis, the halo model is useful since it is able to model the small scales which are mostly impacted by baryoinic feedback accurately.

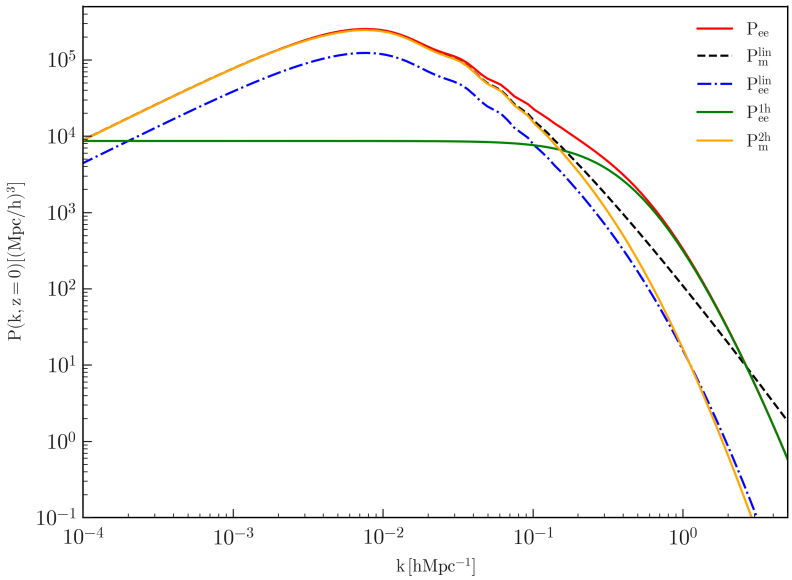



Figure 2.5: A plot showing the gas linear power spectrum, the one-halo, the two-halo, and the non-linear power spectrum, and the linear matter power spectrum. For small $k$, the two-halo term dominates the power spectrum, while for large $k$, the one-halo term dominates the power spectrum. In calculating the non-linear power spectrum, the one-halo term is supressed on larger large scales to remove the contribution of shot noise arising from assuming that halos are infinitely small point on larger scales. The power spectrum is obtained using the halo model formalism described in this section, and the halo profile is obtained using the baryonfication correction model described in section 2.8. I use the cosmological parameters listed in table 3.1. The gas power spectrum converges to the linear matter power spectrum on large scales is modified on small scales due to the impact of baryonic feedback.

## 2.7 Weak gravitational Lensing & cosmic shear

This section is based on the lecture notes of the *Gravitational Lensing* cousrse taught by Prof. Dr. Peter Schneider at the University of Bonn.

Gravitational lensing, in the language of general relativity, is the modification of light (null) geodesics by virtue of the existance of massive objects. In the language of special relativity, it is the bening of light by virtute of the gravitional force of a massive object on a photon. As a result, the image of a bright object (galaxy or a quasar for example) gets magnified and sheared as compared to a situation where there are no lensing effects. In this thesis, I am only interested with the weak field limit of the gravitional lensing where only shear effects are observable. Fig. 2.6 shows a schematic representation of shearing and magnification effects.

In the weak field limit, the shearing effects of gravitational lensing are dominant over the magnification effects. Since the matter density contrast in the present epoch is of order $\delta \approx 1$, it is possible to invoke the weak field approximation to study the shearing effects of distant galaxies' images by the large scale stucture. This effect is called **cosmic shear**.

To understand the lensing of a galaxy image, the shape of a galaxy, or the ellipticity, is represented as

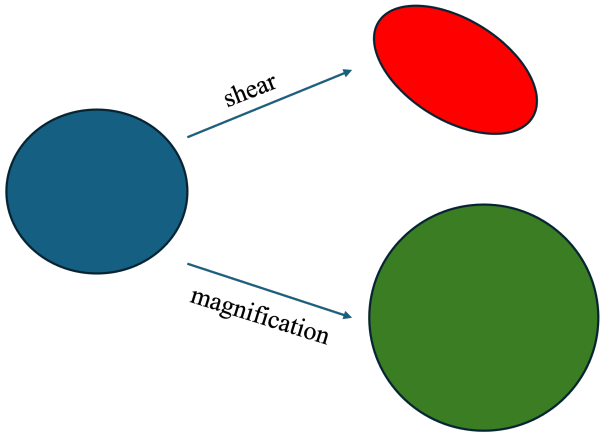



Figure 2.6: A schematic representation of the imapct of gravitational lensing on a source with a circular shape (blue). Lensing has two main effect: shearing (red) and magnification (green). In the weak field limit, the magnificaiton effects are negligible and shearing effects dominate.

a spin-2 field with two components[9] on the unit sphere.

In this section, the symbole $\boldsymbol{\theta}$ is used to indicate a solid angle where $\boldsymbol{\theta} = \theta_1 + i\theta_2$ in the source plane. Fig. 2.7 shows how In practice,$\boldsymbol{\theta}$ is related to a displacement vector $\boldsymbol{\zeta}$ via the relation

$$\boldsymbol{\zeta} = \chi\boldsymbol{\theta}. \tag{2.73}$$

This is clear from Fig. 2.7

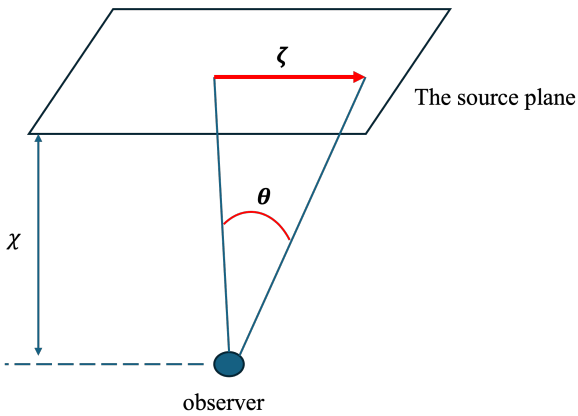



Figure 2.7: A schematic representation showing how the angle $\boldsymbol{\theta}$ is related

The source galaxy shape $\alpha$ is observed with a lensed shape $\beta$ via a transformation $\mathcal{A}$ such that

$$\beta = \mathcal{A}\alpha, \tag{2.74}$$

with $\mathcal{A}$ beging a $2 \times 2$ matrix. $\mathcal{A}$ can be written in terms of two components: the shear $\gamma$ and the

[9] The image of a galaxy has an eliptical shape, therefore, it is possible to describe the shape with two principal components representing the minor axis and the major axis.

convergence $\kappa$ such that

$$\mathcal{A} = \begin{pmatrix} 1-\kappa-\gamma_1 & -\gamma_2 \\ -\gamma_2 & 1-\kappa+\gamma_1 \end{pmatrix}, \tag{2.75}$$

where $\gamma = \gamma_1 + i\gamma_2$ in the the plain of complex numbers. Each of the convergence and the shear has both a geometric and a physical meaning. The trace of the $\mathcal{A}$ is $\mathrm{tr}(\mathcal{A}) = 2(1-\kappa)$, meaning that the magnification effects are controlled by the quatity $\kappa$. Meanwhile, $\gamma$ is the qunatity that controls the shearing of an image. In this formalism, THe convergence $\kappa$ is actullay a field, since it is related to the matter denstiy contrast $\delta$ via the relationship

$$\begin{aligned} \kappa(\boldsymbol{\theta}) &= \int \mathrm{d}\chi\, p_\chi(\chi)\, \kappa(\boldsymbol{\theta}, \chi) \\ &= \frac{3H_0^2\Omega_\mathrm{m}}{2c^2} \int_0^{\chi_\mathrm{h}} \mathrm{d}\chi\, g(\chi)\, f_K(\chi)\, \frac{\delta\big(f_K(\chi)\boldsymbol{\theta}, \chi\big)}{a(\chi)}\,, \end{aligned} \tag{2.76}$$

where $\chi$ is the comoving distance, $f_K(\chi) = \chi$ for a flat universe, and $g(\chi)$ is a geometric factor. Furthermore, $p_\chi(\chi)$ is the source distribution i.e. the probability of findingg a galaxy at cosmoving distance $\chi$.

The shear in turn, is a spin-2 field which is related to the convergence via the relationship

$$\begin{aligned} \gamma(\boldsymbol{\theta}) &= \frac{1}{\pi} \int_{\mathbb{R}^2} \mathrm{d}^2\theta'\, \mathcal{D}(\boldsymbol{\theta} - \boldsymbol{\theta}')\, \kappa(\boldsymbol{\theta}')\,, \quad \text{with} \\ \mathcal{D}(\boldsymbol{\theta}) &\equiv -\frac{\theta_1^2 - \theta_2^2 + 2\mathrm{i}\theta_1\theta_2}{|\boldsymbol{\theta}|^4} = \frac{-1}{(\theta_1 - \mathrm{i}\theta_2)^2}\,. \end{aligned} \tag{2.77}$$

As can be seen from Eq. 2.77, the shear is a convolution of $\kappa$ with the kernel $\mathcal{D}$, or in other words $\mathcal{D}$ is the shear generated by a point mass (if $\kappa$ is a delta function then the shear is equal to $\mathcal{D}$ after the integral). This relation can be inverted to find that

$$\begin{aligned} \kappa(\boldsymbol{\theta}) - \kappa_0 &= \frac{1}{\pi} \int_{\mathbb{R}^2} \mathrm{d}^2\theta'\, \mathcal{D}^*(\boldsymbol{\theta} - \boldsymbol{\theta}')\, \gamma(\boldsymbol{\theta}') \\ &= \frac{1}{\pi} \int_{\mathbb{R}^2} \mathrm{d}^2\theta'\, \mathcal{R}\mathrm{e}\left[\mathcal{D}^*(\boldsymbol{\theta} - \boldsymbol{\theta}')\, \gamma(\boldsymbol{\theta}')\right]\,, \end{aligned} \tag{2.78}$$

where $\kappa_0$ is a constant that arises since $\mathcal{D}$ is not defined at $\boldsymbol{\theta} = 0$. In simple words, shear and convergence are complementary quantities. Since the convergence is related to the matter density along a certain direction $\boldsymbol{\theta}$, so as the shear. Therefore, if the impact of shearing on galaxy images is studied, it is possible to constrain the matter density in the universe.

To relate the shear to the shape of galaxy there is one important assumptions. Galaxies in the universe are randomly oriented, however by virtue of the cosmological principles, the expectation value of galaxy shapes should be zero. If $\epsilon_s$ is the intrinsic shape of galaxy, therefore

$$\langle \epsilon_s \rangle = 0. \tag{2.79}$$

However, the observed shape of a galaxy $\epsilon$ is subject to lensing effects. In the weak field limit, the

expectation value of the observed galaxy shapes is an unbiased estimator of the shear, meaning

$$\langle \epsilon \rangle = \gamma. \tag{2.80}$$

This is an extremely important result which signals the possibility of constraining the matter distribution in the universe via cosmic shear. To use cosmic shear to infere the cosmological parameter, the relationship between the matter power spectrum and the convegence power spectrum is used. This relation is given in the Limber approximation

$$P_{\kappa\kappa}(\ell) = \frac{9H_0^4\Omega_{\rm m}^2}{4c^4}\int_0^{\chi_{\rm h}} {\rm d}\chi\, \frac{g^2(\chi)}{a^2(\chi)}\, P\left(\frac{\ell}{f_K(\chi)}, \chi\right), \tag{2.81}$$

where $P = P(k, \chi)$ is the matter power spectrum at scale $k$ and cosmocving distance $\chi$. Furthermore, the shear power spectrum and the convergence power spectrum are the same, and by virture of Eq. 2.80, the same as the galaxy shapes power spectrum,

$$P_{\kappa\kappa} = P_{\gamma\gamma} = P_{\epsilon\epsilon}. \tag{2.82}$$

Eq. 2.81 shows that any approximation to obtain a theoretical prediction of the matter power spectrum $P(k, \chi)$ will also influence the shear power spectrum $P_{\gamma\gamma}$. One of the main assumptions in this calculation is that structure growth in the late universe is only gravitionally driven. However, baryonic feedback on the large scales structure can influence the distributino of matter in the unvierse on different scales (Daalen et al., 2011). In section 2.8, I discuss how baryonic feedback influences the power spectrum.

## 2.8 The impact of baryonic feedback on the large-scale structure

This section is a qualitative discussion of baryonic feedback on the large sacles structure. It is mainly based on Daalen et al. (2011), one of the first papers to compare and contrast the impact of baryonic feedback on the matter power spectrum. Baryonic feedback are the set of baryonic physics processes that impact the distribution of matter (Rudd, Zentner and Kravtsov, 2008). The real impact of these processes on the matte distributino in the universe is realised by analyzing hydrodynamical simulations. I list below of the feedback mechanism.

- **Active galactic nuclei (AGN) feedback:** AGN feedback originates when gas accretes on a black hole at the center of galaxy and falls into the black hole (Ishibashi and Fabian, 2012). The blackhole then injects energy back into the gas that is accreting, warming up the gas and results in hot jets that can be observed up to scales of $\sim$ 1Mpc. When gas gets pushed ouside the perimeter of a galaxy, it creates a potential well that matter follows, reducing the clustering of matter inside a galaxy. Therefore, the impact of AGN feedback is manifested as supression of power at scales of order $k \sim 10\text{h.Mpc}^{-1}$ compared to a scenario where only gravity influences structure fomation in the unvierse (Martin-Alvarez et al., 2025).

- **Supernova feedback:** supernova feedback is a sudden collapse large-mass stars under the influence of gravity, leading to the release of energies in form of photons and charged particles on the order

of $\sim 10^{51}$erg[10] (S. Huang et al., 2020). The energy reeleased warms up the gas within a galaxy and and enriches the interstellar medium and the circumgalactic medium with heavy metals that formed throughout the life time of the star. The exact impact on the matter power spectrum is not known a priori since each scenario of supernova feedback leads to different results and affect a range of (small) scales, and there is not reason to believe one scenario over the other. Nonetheles, by studying the different possible scenario, the supernova feedback on cosmological scales leads to $\sim 1 - 10\%$ suppression in the matter power spectrum on scales between $k \sim 1 - 10\text{h.Mpc}^{-1}$ (Daalen et al., 2011).

- Star formation feedback: Due to multiple cooling mechanisms the gas in galaxy collapses under the influence of gravity to form stars. Star formation leads to increase of power up to $\sim 10\%$ on scales $k \gtrsim 10\text{h.Mpc}^{-1}$ (Daalen et al., 2011).

Fig. 2.8 shows a plot comparing the ratio of the power spectrum of dark matter only scenario $P_\delta^{\text{DMO}}$ to the matter power spectrum in different hydrodynamical simualtions $P_\delta^{\text{hydro}}$. Each of the given hydrodynamical simulations has different implmentation of small scale physics and is parameterized by a set of nuisance paremters of no particular cosmological. Using astrophysical observations, it is not possible to probe one of these scenarios over the other. Therefore, to mitigate the impact of baryonic feedback on the large scale structure, multiple statistical tools have been introduced, as discussed in the introduction. Other analytical approches involved parametrizing baryonic feedback with analytical models calibrated on hydrodynamical simulations (Mead et al., 2021b; A. Schneider and Teyssier, 2015). These models have two shortcomings. First, the hydrodynamical simulation used for calibration is not guarenteed to agree with the true scenario of baryonic feedback in the universe. Second, the parameters that goven baryonic feedback have uniform and wide prior range, meaning that the constraining power of cosmological data will be absorbed by these nuisance parameters, leaving the cosmological parameters with no constraining power (A. Schneider, Kovač et al., 2025).

To obtain tight priors on the baryonic feedback parameter, it is possible to use probes of gas distribution in the Universe which in turns traces the baryonic matter distribution. This can be done by observing the kinematic Sunyaev–Zeldovich (kSZ) effect from CMB maps (Sunyaev and Zeldovich, 1972; Liu et al., 2025) or the X-ray observation of galaxy clusters . the kSZ is the inverse scattering of CMB photons when interracting with free energetic free electrons due to their bulk motion. This effect is observed as a secondary anisotropy in the observed maps of CMB. It is possible to relate the observed anisotropy $\Delta T_{\text{kSZ}}$ due to kSZ with the electron number density along the line of sight

$$\frac{\Delta T_{\text{kSZ}}}{T_{\text{CMB}}} = \frac{\sigma_{\text{T}}}{c} \int_{\text{los}} \mathrm{d}l \, e^{-\tau} n_{\text{e}} \, v_{\text{p}}, \tag{2.83}$$

where $T_{\text{CMB}} = 2.725\text{K}$ the blackbody temprature of the CMB, $\sigma_{\text{T}}$ the Thomson cross-section, $\tau$ the optical depth, $l$ is the co-moving distance along the line of sight, and $v_{\text{p}}$ is the peculiar velocity of gas. Eq. 2.83 indicates that it is possible to proble the distribution of baryons in the universe by extracting the secondary anisotropies in the CMB that are caused by the kSZ effect. Combining the measurements of kSZ with constraints on the hot gas mass fraction from X-ray observations of galaxy clusters can impose constraints constraints on the baryonic feedback parameters. These constraints can be used as priors in a comsological analysis of cosmic shear data yielding tighter constraints on the comsological parameter as compared to having wide priors on these paraneters.

[10] 1 Joul = $10^7$ erg

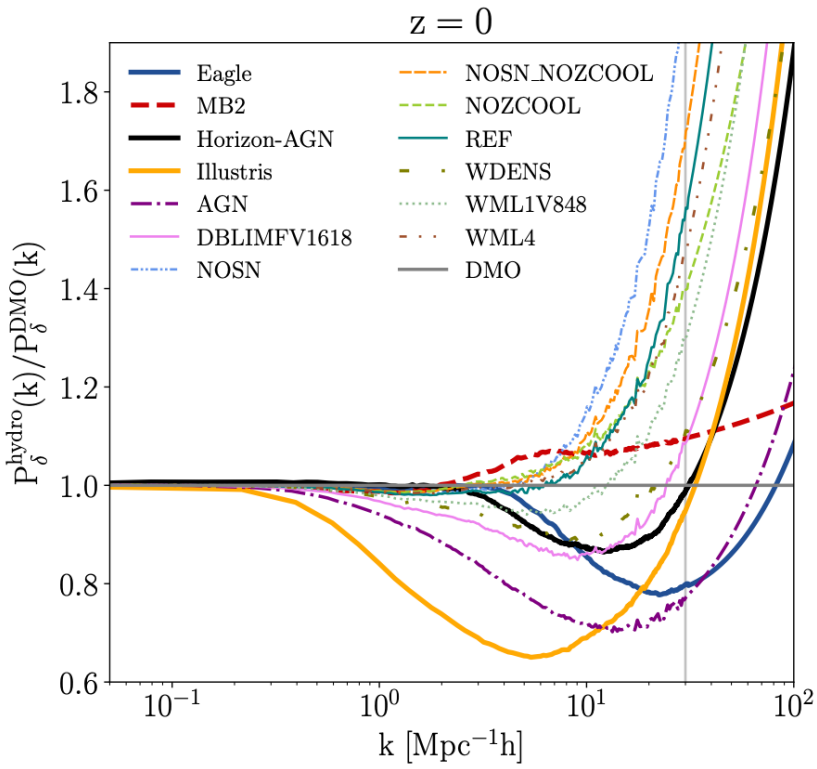



Figure 2.8: A plot adopted from H.-J. Huang et al. (2019) showing the supression of the power spcctrum for different hydrodynamical simulations with different models for baryonic feedback. For simulations where supernova feedback and AGN feedback are present, the power is superessed on the scales where the gas pressure from these two feedback mechanism influences matter distribution. The ratio rises for larger for smaller scales since gas cooling leads to star formation, boosting the power on small scales.

Though kSZ is a promising tool, extracting its signal from CMB observation requires assumptions about the temprature and ionization of the gas in galaxy clusters and the bulk velocity of the clusters (Carlstrom, Holder and Reese, 2002). Thankfully, the measured DM of FRBs is a direct probe of the electron number denisty in the unvierse and thereofre baryonic feedback. In section 2.9, I describe the use of FRBs in cosmology in addition to the systematic and measurement uncertainties which impacts the modelling and measurements of the DM signal.

## 2.9 Fast radio bursts

This section is based on Alonso (2021b), Reischke and Hagstotz (2023) and Petroff, J. W. Hessels and Lorimer (2019). For ease of writing, I consider that $\hbar = c = 1$ in this section. As mentioned in the introduction, the DM of FRBs is proportional to the difference in arriveal time between high frequency wave and small frequency wave, and I show how this relationship rises below. Since the FRBs passes through the intergalactic medium (IGM), it interacts with the plasma in this medium. The plasma is made up out of charged electrons. Therefore, when a light wave of frequency $\omega$ and wave vector $\mathbf{k}$, the

plasma follows the dispersion relation

$$\omega^2 - |\mathbf{k}|^2 = \omega_e^2, \quad \omega_e \equiv \frac{4\pi e^2 n_\mathrm{e}}{m_e}, \tag{2.84}$$

where $n_\mathrm{e}$ is the electron number density as measured by the plasma's frame. The velocity of the electromagnetic wave packet in the plasma's frame of reference $v_g$, is equivalent to the velocity of a massive particle with mass $\omega_e$ and energy $\omega$

$$v_g = \sqrt{1 - \frac{\omega_e^2}{\omega^2}}. \tag{2.85}$$

Since the FRBs are in the radio regime, the ratio $\frac{\omega_\mathrm{e}^2}{\omega^2} << 1$, and the group velocity can be approximated assume

$$v_g \approx \left(1 - \frac{1}{2}\frac{\omega_\mathrm{e}^2}{\omega^2}\right). \tag{2.86}$$

To calculate the time delay due to scattering of the free electrons, I turn the problem into calculating the time daley between the arrival of a pulse with frequency $\nu_0$ and a pulse of infinite frequency. I call these pulses a normal and infinite pulses respectively. Once I have an expression for the time delay, it possible to calculate the time delay between pulses of a high frequency $\nu_\mathrm{high}$ and a low frequency $\nu_\mathrm{low}$ Theoretically, plasma does not affect the infinite pulse since the pulse has infinite momentum. Therefore, the group velocity of the infinite pulse is 1. I use the setup in Fig. 2.9 to find an expression of the imte delay. The setup shows freely falling frame of references along the trajectory of the normal pulse. The time delay beetween the arrival of the normal pulse and the arrival of the infinite pulse (the two events are marked by an **x** on the yellow lines) can be given by

$$\mathrm{d}\tau = \frac{\mathrm{d}l}{v_g} - \mathrm{d}l, \tag{2.87}$$

with $\tau$ being the proper time measured by the plasma (notice that I used the group velocity relative to the plasma) and $\mathrm{d}l$ the proper ditsnce between the two stations. The correponding time delay measured by the comoving observer $\mathrm{d}t$ is given by

$$\mathrm{d}t = (1 + z)\mathrm{d}\tau. \tag{2.88}$$

If the normal pulse has a 4-momentum defined as $P^\mu = \mathrm{d}x^\mu/\mathrm{d}\Lambda$ (see Eq. 2.14), where $\lambda$ is an affine parameter, the proper length can be written as $\mathrm{d}l = P^\mu u_{e,\mu}$, where $u_{e,\mu}$ is the station's 4-velocity as measured by the observed. Thus to lowest order in $\omega_\mathrm{e}/\omega$, the measured time $\mathrm{d}t$ is written

$$\mathrm{d}t = \frac{(1+z)\omega_\mathrm{e}^2}{2\omega^2}\left(P^\mu u_{e,\mu}\right)\mathrm{d}\lambda. \tag{2.89}$$

Finally, due to cosmological redshift, the station observes a frequency $\omega$ different from the what the observers detects, as shown by Eq. 2.28. We relate the two frequencies via the relation $\omega = 2\pi\nu_0(1 + z)$,

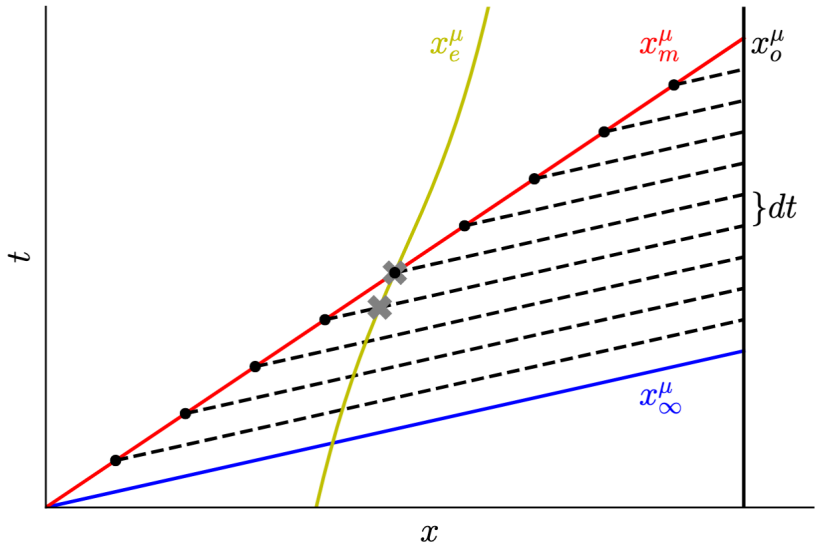



Figure 2.9: Spacetime diagram showing the evaluation of time dalay between a pulse of frequency $\nu_0$ (world line in red) and a pulse of infinite frequency (world lines in blue and dashed). The plasma particles follow world lines given in yellow color. The figure is adopted from Alonso (2021b)

.

making the final expression for the time delay

$$\Delta t = \frac{e^2}{2\pi m_{\rm e} \nu_o^2} \mathrm{DM}, \tag{2.90}$$

where DM is the dispersion measure given as an integral

$$\mathrm{DM} \equiv \int_{\lambda_{\rm e}}^{\lambda_o} \mathrm{d}\lambda \left( n_{\rm e} \frac{P^\mu u_{e,\mu}}{1+z} \right) \Bigg|_\lambda , \tag{2.91}$$

where the subscript $\lambda$ is a reminder tha tthe integrand is evaluated along the pulse's trajectory. In a realistic situation, the plasma has a peculiar velocity and the normal pulse does not follow a null geodesic. I consider a situation where the stations are comoving observer with 4-velocity $u_e^\mu = (1, 0, 0, 0)$, and the pulse moves along a null geodesic in an unperturbed universe with momentum $P^\mu = (A/a, A/a^2, 0, 0)$[11], where $A$ is a constant. The inner producet of momentum and veloctiy reduces to

$$P^\mu u_{e,\mu} = g_{tt}(A/a)(1) = -A/a. \tag{2.92}$$

[11] I omit the proof to obtain this results. The proof can be done by solving the geodesic equation (Eq. 2.2) and setting the coordinate origin such that the photon follows a geodesic with $\phi = 0$ and $\theta = 0$.

Since $\lambda$ parametrizes the geodesic of the pulse, it is possible to change variables in Eq. 2.91 using the expression for the 4-momentum where $\mathrm{d}t/\mathrm{d}\lambda = A/a$. The integral becomes

$$\mathrm{DM} \equiv \int_{t_o}^{t_e} \mathrm{d}t\, a\, n_\mathrm{e}, \tag{2.93}$$

where I have changed the integral pounds. Remeber that $n_\mathrm{e}$ is the physical density as measured by the plasma. It can be written in terms of the comoving density as $n_\mathrm{e} = n_{\mathrm{e},c} a^{-3}$. Therefore, the correpoinding comoving density scales with Using the relationship between cosmic time $t$ and the sacle factor $a$ as in Eq. 2.26 and the relationship between the $a$ and $z$ in Eq. 2.29, the integral can be written as

$$\mathrm{DM} \equiv \int_0^{z_e} \mathrm{d}z' \frac{1+z'}{H(z')} n_{\mathrm{e},c}(z'). \tag{2.94}$$

Hereafter, I work with comiving qunatities and will label the comoving electorn density at a redshift $z$ with $n_\mathrm{e}$. Eq. 2.94 is the dispersion measure-redshift relation $\mathrm{DM}_z$ which I use in this thesis to study the impact of systematic uncertainties. Eq. 2.84 includes an expression of the comoving electron number density $n_{\mathrm{e},c}$ this quantity can be written in terms of the baryon density fraciton $\Omega_\mathrm{b}$ given some consideration to the ratio of free electron in the universe that disperses the signal. In general, the expression should look like

$$n_\mathrm{e} = \frac{\bar{\rho}_\mathrm{b}}{m_\mathrm{p}} F(z), \tag{2.95}$$

where $\bar{\rho}_\mathrm{b}$ is the average of the baryonic density in the universe, $m_\mathrm{p}$ is the mass of proton, and $F(z)$ is the fraciotn of free electrons in the intergalactic medium. To find $F(z)$, I make the following considerations

- If the universe has only helium and hydrogen atom, then the fraction of free electrons in the universe is $\chi_\mathrm{e} = Y_\mathrm{H} X_{\mathrm{e,H}}(z) + \frac{1}{2} Y_\mathrm{H} X_{\mathrm{e,He}}(z)$, with $Y_\mathrm{H}$, $Y_\mathrm{He}$, $X_{\mathrm{e,H}}(z)$, and $X_{\mathrm{e,He}}(z)$ the mass fraction of hydrogen, the mass fraction of helium, the ionized fraction of hydrogen, and the ionized fraction of helium respectively. I multiply $Y_\mathrm{He}$ with a factor $1/2$ since the helium atom has two the Using big bang nucleosyntheis, I assume that molecules in the unvierse are in the form of either hydron or helium with the fraction of hydrogen $Y_\mathrm{H} = 0.75$ and the fraction of helium is $Y_\mathrm{He} = 0.25$.
- Hydrogn and helium are fully ionized for $z \lesssim 3$ (Meiksin, 2009), meaning that the ionized fraction of Hydrogen $X_{\mathrm{e,H}}(z)$ and $X_{\mathrm{e,He}}(z)$ are both equal to 1.
- The fraction of free electrons in the intergalactic medium $f_\mathrm{IGM}$ is constant for the range of redshifts considered in this problemfind a citations for this.

By dropping the redshift dependence on the ionized fraction, I get an expression of $F(z)$ that is constant in redshift for the range of redshifts considered in this thesis,

$$F(z) = f_\mathrm{IGM} \left[ Y_\mathrm{H} X_{\mathrm{e,H}} + \frac{1}{2} Y_\mathrm{He} X_{\mathrm{e,He}} \right], \tag{2.96}$$

Writing $\bar{\rho}_\mathrm{b} = \Omega_\mathrm{b} \rho_\mathrm{cr}$, The final form of Eq. 2.94 is given by

$$\mathrm{DM}_\mathrm{LSS}(z) = \mathcal{A} \int_0^z \frac{1+z'}{E(z')} \mathrm{d}z', \tag{2.97}$$

with $\mathcal{A} := \frac{3c\Omega_{b0}H_0}{8\pi G m_p}\chi_e f_{IGM}$, and $E(z) = H(z)/H_0$. This is the expression for the DM that rises from free electrons that follows the large scales structure, hence, I will label the DM that rises from the cosmic electron density with $DM_{LSS}$ hereafter. As seen from 2.97, $DM_{LSS}$ is highly sensitive to he baryon density in the universe $\Omega_b$ and the Hubble constant $H_0$.

Eq. 2.97 expresses the mean DM at a redshift $z$. For a perturbed universe, the electron number density field is also a function of the angular direction $\hat{\mathbf{x}}$. I defined the electron number density constrast $\delta_e(\hat{\mathbf{x}}, z)$ in terms of the mean electorn density $n_e(z)$ and the electron density in a perturbed universe $n_e(\hat{\mathbf{x}}, z)$

$$\delta_e(\hat{\mathbf{x}}, z) = \frac{n_e(\hat{\mathbf{x}}, z) - n_e(z)}{n_e(z)}. \tag{2.98}$$

The corresponding $DM_{LSS}$ is therefore

$$DM_{LSS}(\hat{\mathbf{x}}, z) = \mathcal{A}\int_0^z \frac{1+z'}{E(z')}(1+\delta_e(\hat{\mathbf{x}}, z'))dz' . \tag{2.99}$$

It is clear that $DM_{LSS}(\hat{\mathbf{x}}, z)$ is a random field since it is related to $\delta_e(\hat{\mathbf{x}}, z')$ which in turn is related to the matter density contrast $\delta$.

Using the formalism introduced in sections 2.5 and 2.6, it is possible to use the electron power spectrum $P_{ee}$ to express the correlation between two measured $DM_{LSS}$ at a angular positions $\left(\hat{\mathbf{x}}_i, \hat{\mathbf{x}}_j\right)$ and at redshifts $(z_i, z_j)$, namely the covariance matrix. (Reischke and Hagstotz, 2023) showed that the cosmological covariance of $DM_{LSS}$ is derivd by first writing the correlator bracket

$$\mathrm{cov}_{ij} = \langle DM_{LSS}(\hat{\mathbf{x}}_i, z_i) DM_{LSS}(\hat{\mathbf{x}}_j, z_j)\rangle - DM_{LSS}(z_i) DM_{LSS}(z_j), \tag{2.100}$$

Simplifying, the expression reduced to

$$\mathrm{cov}_{ij} = \int_0^{z_i} dz_i' \, W^i_{DM}(z_i') \int_0^{z_j} dz_j' \, W^j_{DM}(z_j') \left\langle \delta_e(\hat{\mathbf{x}}_i, z_i')\delta_e(\hat{\mathbf{x}}_j, z_j')\right\rangle , \tag{2.101}$$

where I have defined the weight $W_{DM}(z)$ as

$$W_{DM}(z) = \begin{cases} \mathcal{A}\dfrac{1+z}{E(z)} & z < z_{FRB} \\ 0 & z > z_{FRB} \end{cases}, \tag{2.102}$$

with $z_{FRB} = z_i, z_j$ the redshift of an FRB. The electron correlator bracket in real space can be written in harmonic space such that

$$\left\langle \delta_e(\hat{\mathbf{x}}_i, z_i)\delta_e(\hat{\mathbf{x}}_j, z_j)\right\rangle = \int \frac{d^3k}{(2\pi)^3} e^{i\mathbf{k}\cdot(\mathbf{x}_i - \mathbf{x}_j)} P_{ee}(k, z_i, z_j), \tag{2.103}$$

where I define the power spectrum in Fourier space in similar fashion to 2.59 as

$$\langle \delta_e(\mathbf{k}, z_i)\delta_u^*(\mathbf{k}', z_j)\rangle = (2\pi)^3 P_{uu}(k, z_i, z_j)\delta_D(\mathbf{k} - \mathbf{k}'), \tag{2.104}$$

and $\mathbf{x}$ is the cosmoving spatial position and related to the comoving distance as $\mathbf{x} = (\chi\hat{\mathbf{x}}, \chi)$.

To perform the integral in 2.103, I work in spherical coordinate sin the $k$-space, and decompose the integral into the radiu $k$ and a solid angle $\Omega_k$. In addition, I expand the exponential $e^{i\mathbf{k}\cdot(\mathbf{x}_i-\mathbf{x}_j)}$ into plane waves (As explained in the appendix)

$$\begin{aligned}
&\left\langle \delta_{\rm e}(\hat{x}_i, z_i)\delta_{\rm e}(\hat{x}_j, z_j)\right\rangle \\
&= \frac{2}{\pi}\int k^2 {\rm d}k \int {\rm d}\Omega_k\, P_{\rm ee}(k, z_i, z_j) \\
&\quad \times \sum_{\ell,\ell'}\sum_{m,m'} {\rm i}^{\ell}(-{\rm i})^{\ell'} Y_{\ell m}(\hat{k})\, Y^*_{\ell m}(\hat{x}_i)\, j_\ell(k\chi_i)\, Y^*_{\ell' m'}(\hat{k})\, Y_{\ell' m'}(\hat{x}_j)\, j_{\ell'}(k\chi_j) \\
&= \frac{2}{\pi}\int k^2 {\rm d}k\, P_{\rm ee}(k, z_i, z_j) \sum_\ell \sum_m Y^*_{\ell m}(\hat{x}_i)\, j_\ell(k\chi_i)\, Y_{\ell m}(\hat{x}_j)\, j_\ell(k\chi_j) \\
&= \frac{1}{2\pi^2}\sum_\ell (2\ell+1)\int k^2 {\rm d}k\, P_{\rm ee}(k, z_i, z_j)\, j_\ell(k\chi_i)\, j_\ell(k\chi_j)\, P_\ell(\cos\theta),
\end{aligned} \tag{2.105}$$

with $Y_{\ell m}$ and $j_\ell$ being the spherical harmonic function and the the spherical Bessel function of the first kind respectively. To simplify the sum, I use the property

$$\sum_m Y_{\ell m}(\hat{\mathbf{x}}_i) Y^*_{\ell m}(\hat{\mathbf{x}}_j) = \frac{2\ell+1}{4\pi} P_\ell(\cos\theta), \tag{2.106}$$

with $\cos\theta_{ij} = \hat{\mathbf{x}}_i.\hat{\mathbf{x}}_j$ and $P_\ell$ is the Legendre polynomial of degree $\ell$ (defined in the appendix). In addition, the power spectrum at two as a funciton of two redshifts $z_i$ and $z_j$ can be replaced by its geometric mean such that $P_{\rm ee}(k, z_i, z_j) = \sqrt{P_{\rm ee}(k, z_i) P_{\rm ee}(k, z_j)}$ (Kitching and Heavens, 2017). The final expression for the cosmological covariance of $\mathrm{DM_{LSS}}$ is

$$\begin{aligned}
{\rm cov}_{ij}(\cos\theta, z_i, z_j) &= \frac{1}{2\pi^2}\sum_\ell (2\ell+1) P_\ell(\hat{x}_i\cdot\hat{x}_j)\int k^2 {\rm d}k \\
&\quad \times \int_0^{z_i} {\rm d}z'_i\, W^i_{\rm DM}(z'_i)\sqrt{P_{\rm ee}(k, z'_i)}\, j_\ell(k\chi_i) \\
&\quad \times \int_0^{z_j} {\rm d}z'_j\, W^j_{\rm DM}(z'_j)\sqrt{P_{\rm ee}(k, z'_j)}\, j_\ell(k\chi_j) \\
&= \sum_\ell \frac{2\ell+1}{4\pi} P_\ell(\cos\theta_{ij}) C_{ij}(\ell).
\end{aligned} \tag{2.107}$$

The expression contains two integral where one integral goes over the sacle $k$ and two integrals to include the $\mathrm{DM_{LSS}}$ contribution from redshift $z = 0$ up to the redshift of the FRB $z = z_i,\, z_j$. In this thesis, fast calculation of the covariance matrix is essential since I use a compression step to pass one simulation instance in the simulation based inference pipeline. Therefore, I use the Limber approximation (Limber, 1953) to accelerate the calculation of the covaraince. I show a comparison between the Limber

approximation and the non-Limber approximation in section 3.1.3. The Limber approximation assumes tha the spherical bessel function $j_\ell$ can be effectively replaced by a Dirac-Delta distribution such that

$$j_\ell(k\chi) \to \sqrt{\frac{\pi}{2(\ell+\frac{1}{2})}}\,\delta_D((\ell+\frac{1}{2})-k\chi). \tag{2.108}$$

The word *effectively* is important in this context since the $j_\ell$ by itself is **not** approximated by a the expression in Eq. 2.108, however, the two resultant integrals are equivilant in some regimes of $\ell$. Invoking, the approximation in the full expression of the covariance and changing the integration vaiable into the comoving radius $\chi$, I obtain the final expression

$$C_{ij}(\ell) = \int_0^{\min(\chi_i,\chi_j)} \frac{\mathrm{d}\chi}{\chi^2}\frac{\mathrm{d}z}{\mathrm{d}\chi}\frac{\mathrm{d}z}{\mathrm{d}\chi}\, W^j_{\mathrm{DM}}(z(\chi))\, W^i_{\mathrm{DM}}(z(\chi))\, P_{\mathrm{ee}}\left(\frac{\ell+1/2}{\chi}, z'(\chi)\right). \tag{2.109}$$

The choice of the Limber approximation is valid when the terms multiplied by $j_\ell$ are slowly changing realtive to $j_\ell$. The calculation via the Limber approximation agrees with the brute force calcualtion for large $\ell$ i.e the regime where $j_\ell$ oscillates very quickly. This is evident by noticing that error on the Limber approximation scales with $O(\ell^{\frac{1}{2}})$.

Eq. 2.107 has another bottleneck. The sum over the Legendre polynomials is another computationally expensive step that scales as $O(\ell)$. At small enough angular separations, it is possible to approximate the spherical coordinates as flat, in what is called the flat sky approximations (Kaiser, 1992). In this approximation, the relationship between the angular power spectrum is derived by taking the Fourier tansform in two-dimesions of the field instead of the spherical harmonic transform. Starting from the last line of Eq. 2.107, the property in Eq. 2.106, and the definition of the angular power spectrum in Eq. 2.64, the covariance can be re-written as[12]

$$\mathrm{cov}_{ij} = \langle \mathrm{DM}(\hat{\mathbf{x}}_i, z_i)\mathrm{DM}(\hat{\mathbf{x}}_j, z_j)\rangle, \tag{2.110}$$

where I used

$$\langle \mathrm{DM}_{\ell m}(z_i)\mathrm{DM}_{\ell' m'}(z_j)\rangle = \delta_{\ell\ell'}\delta_{mm'}C_{ij}(\ell), \tag{2.111}$$

and

$$\mathrm{DM}_{\ell m}(z_i) = \sum_{\ell m} Y_{\ell m}(\hat{\mathbf{x}}_i)\mathrm{DM}_{\ell m}(z_i). \tag{2.112}$$

To invoke the flat sky approximation, I invoke the Fourier transform of $\mathrm{DM}(\hat{\mathbf{x}}_i, z_i)$ in Eq. 2.110 and assume that in the two-dimensional Fourier space

$$\langle \mathrm{DM}(\ell, z_i)\mathrm{DM}(\ell', z_j)\rangle = (2\pi)^2\,\delta^D(\ell-\ell')C_{ij}(\ell), \tag{2.113}$$

I end up with the following expression

$$\mathrm{cov}_{ij} = \int \mathrm{d}\boldsymbol{\ell}\, e^{-i\boldsymbol{\ell}.\boldsymbol{\theta}_{ij}} C_{ij}(\ell), \tag{2.114}$$

with volume element can be written as $\mathrm{d}\boldsymbol{\ell} = \ell\mathrm{d}\ell\mathrm{d}\alpha$, and the integral can be performed in the angular

[12] I omit the subscript from $\mathrm{DM}_{\mathrm{LSS}}$ hereafter, where the symbol DM refers to the cosmological contributiuon to the DM.

position of the Fourier space $\alpha$. I use the properties of the Bessel function of the first kind

$$J_0(\ell\theta_{ij}) = \frac{1}{2\pi}\int_0^{2\pi} e^{i\ell\theta_{ij}\cos\alpha}\,\mathrm{d}\alpha, \tag{2.115}$$

The covariance matrix in the flat sky approximation is

$$\mathrm{cov}_{ij} = \int \frac{\ell\mathrm{d}\ell}{2\pi} J_0(\ell\theta_{ij})\, C_{ij}(\ell). \tag{2.116}$$

This result is very useful to further accelerate the calculation of the covariance matrix. Though only valid on small angular separation, this apprixomation is widely used for forecasting purposes. Simulating the dispersion to fulfill the puprpose of this thesis will be described in section 3.2.

Now that an expression of the covariance is obtained, I focus on deriving an expression for the variance of the dispersion measure signal at a certain redshift $z$. The variance of the DM sinal is defined as

$$\sigma^2\left[\mathrm{DM}(\hat{\mathbf{x}}, z)\right] = \langle\left((\mathrm{DM}(\hat{\mathbf{x}}, z) - \mathrm{DM}(z))^2\right)\rangle. \tag{2.117}$$

The expression can be easily derived from Eq. 2.107 by setting $\hat{\mathbf{x}}_i = \hat{\mathbf{x}}_j$ and $z_i = z_j$ writing $C_{ij}(\ell) = C(\ell, z)$, yielding

$$\sigma^2\left[\mathrm{DM}(\hat{\mathbf{x}}, z)\right] = \sum_\ell \frac{2\ell+1}{4\pi} C(\ell, z), \tag{2.118}$$

and if I use the flat sky approximation, the expression changes to

$$\sigma^2\left[\mathrm{DM}(\hat{\mathbf{x}}, z)\right] = \int \frac{\ell\mathrm{d}\ell}{2\pi} C(\ell, z). \tag{2.119}$$

In total, the variance of the DM field is an importnat quantity for constraining baryonic feedback. As you can see from Eq. 2.97, the mean DM-$z$ relation is strongly sensitive to the value of the baryonic density ratio $\Omega_\mathrm{b}$ and the Hubble constant $H_0$ and dependent on the other density ratios through the rate of expansion $E(z)$. Therefore, the dependednce on the baryonic feedback paramerters is entirely encoded in $\sigma^2\left[\mathrm{DM}(\hat{\mathbf{x}}, z)\right]$ and the electron density contrast $\delta_\mathrm{e}$. I will explain how to model the electron power spectrum using an analytical model of the baryonic feedback in section 3.1.1. Fig. 2.10 shows a plot of the dipersion measure redshift relation for different values of $\Omega_\mathrm{b}$ and differnet values for baryonic feedback parameters.

Figure 2.10: A plot showing the impact of varying $\Omega_\mathrm{b}$ on the mean DM-$z$ relation and the impact of varying $\log_{10} M_\mathrm{c}$ one of the baryonic feedback parametrs on on the variance of the DM-$z$ relation.

So far, I have only introduced the cosmological DM. In reality, the measured DM receives contributions from the Milkyway galaxy and the host galaxy contribution which are not included in Eq. 2.97. We label each of these contribution with $\mathrm{DM}_\mathrm{MW}$ and $\mathrm{DM}_\mathrm{host}$ respectively. We discuss these two contributions in section 2.10. The total DM measured as measured on Earth is

$$\mathrm{DM}_\mathrm{tot} = \mathrm{DM}_\mathrm{LSS} + \mathrm{DM}_\mathrm{MW} + \mathrm{DM}_\mathrm{host}. \tag{2.120}$$

describe how modelling the PDF is a hard thing to do To forward model observations, we convolve the $DM - z$ represented by $P(DM|z)$ with measurement and systematic uncertainties, in addition to other observational effects. A catalog of detected FRBs is expected to have observed dispersion measure $DM^i_{\rm obs}$, galactic coordinates, and redshifts for a subset of and observations $z^i$. We assume that each detected FRB was emitted at redshift $z^i$ where the difference between identified and unidentified host can be integrated in how $P(z^i)$ is modeled. Assume that we observe $N_{FRB}$ fast radio bursts signal. We express the observations as the probability of detecting an FRB $DM_{\rm obs}$ given a set of estimated redshifts $\{z^i\}$, can be decomposed using the Bayes rule such that

$$P(DM_{\rm obs}|\{z^i\}) \propto P(\{z^i\}|DM_{\rm obs})P(DM_{\rm obs}). \tag{2.121}$$

Furthermore, if we assume that each observation $z_i$ is independent of other observations we may use the Bayes rule to decompose $P(\{z^i\}|DM_{\rm obs})$ such that

$$P(\{z^i\}|DM_{\rm obs}) = \prod_i P(z^i|DM_{\rm obs}). \tag{2.122}$$

Furthermore, the individual terms in Eq. 2.122 could be derived using the joint probability distribution of the observed and the dispersion measure contribution from the large scale structure, namely $DM_{\rm obs}$ and $DM$ such that

$$\begin{aligned} P(z^i|DM_{\rm obs}) &\propto P(DM_{\rm obs}|z^i)P(z^i) \\ &= P(z^i)\int P(DM_{\rm obs}, DM|z^i)\mathrm{d}DM \\ &= P(z^i)\int P(DM_{\rm obs}|z^i, DM)P(DM|z^i)\mathrm{d}DM. \end{aligned} \tag{2.123}$$

The final expression that relates the probability distribution function of an observation to other modelled terms is

$$\begin{aligned} P(DM_{\rm obs}|\{z^i\}) \propto \\ P(DM_{\rm obs})\prod_i P(z^i)\int P(DM_{\rm obs}|z^i, DM)P(DM|z^i)\mathrm{d}DM. \end{aligned} \tag{2.124}$$

Furthermore, to model each component of the resulting expression, we start with $P(DM_{\rm obs})$. There are three components that contribute additively to $DM_{\rm obs}$: the large scale structure contribution $DM_{\rm LSS}$, the Milkyway contribution $DM_{\rm MW}$, and the host galaxy contribution $DM_{\rm host}$, and the observed dispersion measure is given as

$$DM_{\rm obs} = DM_{\rm LSS} + DM_{\rm MW} + DM_{\rm host}. \tag{2.125}$$

For simplicity, we indicate to the contribution from the large-scale structure simply with $DM$ which is the same in Equation Eq. 2.124. Consider the joint probability distribution $P(DM, DM_{\rm MW}, DM_{\rm host})$, this can be related to $P(DM_{\rm obs}) = P(DM + DM_{\rm MW} + DM_{\rm host})$ via the constrained integral

$$\begin{aligned} P(DM_{\text{obs}}) &= P(DM + DM_{\text{MW}} + DM_{\text{host}}) \\ &= \int_C P(DM, DM_{\text{MW}}, DM_{\text{host}})\, \mathrm{d}DM\, \mathrm{d}DM_{\text{MW}}\, \mathrm{d}DM_{\text{obs}} \\ &= \int_C P(DM)P(DM_{\text{MW}})P(DM_{\text{host}})\, \mathrm{d}DM\, \mathrm{d}DM_{\text{MW}}\, \mathrm{d}DM_{\text{obs}}, \end{aligned} \tag{2.126}$$

Where $C = \{DM, DM_{\text{MW}}, DM_{\text{host}}; DM_{\text{obs}} = DM + DM_{\text{MW}} + DM_{\text{host}}\}$ Similarly, I can model $P(DM_{\text{obs}}|z^i, DM)$ as

$$\begin{aligned} P(DM_{\text{obs}}|z^i, DM) = \int_C & P(DM|z^i, DM)P(DM_{\text{MW}}|z^i, DM) \\ & P(DM_{\text{host}}|z^i, DM)\, \mathrm{d}DM\, \mathrm{d}DM_{\text{MW}}\, \mathrm{d}DM_{\text{obs}}, \end{aligned} \tag{2.127}$$

Each of the terms making up the integral could be further reduced

$$P(DM|z^i, DM) \propto P(DM|z^i)P(DM) \tag{2.128}$$

$$P(DM_{\text{MW}}|z^i, DM) = P(DM_{\text{MW}}) \tag{2.129}$$

$$P(DM_{\text{host}}|z^i, DM) \propto P(DM_{\text{host}}|z^i)P(DM|z^i). \tag{2.130}$$

Keeping in mind, that $DM_{\text{host}}$ that we use in Eq. 2.124 is flat since we do not have any information of the

discuss the motivation for the SBI method.

## 2.10 The systematic uncertainties

In this section, I describe the modelling of systematic uncertainties on the dispersion measure.

- **Milkyway contribution:** The Milkyway (MW) galaxy, just like any disc galaxy, has free electrons in the interstellar medium (ISM). As the FRB pulse pass through the interstellar medium, it gets further dispersed by the free electron in the interstellar medium, adding an additional contribution to the measured $\text{DM}_{\text{tot}}$. There are curently no analytical model that can accuratley describe the electron number density in the MW, however I use the NPE2001 model which is an empirical model that is calibrated on quasars data, and which we will use in this thesis (Ocker and Cordes, 2024). Nonetheless, the model is highly uncertain near the galactic plane meaning that subtracting $\text{DM}_{\text{MW}}$ obtained from this model from $\text{DM}_{tot}$ leads to large measurement error if the FRB source is located near the galactic plane. Fig 2.11 shows a map of the MW contribution to DM as predicted by the NE2001 model.

- **Host galaxy contribution:** FRB picks up a contribution from the galaxy that hosts the FRB source, namely the host galaxy. To model the host galaxy contribtion, N-body simulation provide the best information to date. Using the CAMELS simulation (Villaescusa-Navarro et al., 2021),

Figure 2.11: A map of the MW contribution to the measured DM with the NPE2001 model. The model is calibrated on data from quasars. As seen from the map, $\mathrm{DM}_{\mathrm{MW}}$ is large a the galactic plane and lower towards the pole since baryons in the MW are situated in a disk-like shape.

Theis et al. (2024) showed that the variability of the host contribution with redshift is mainly driven by the variability of star formation with redshift while the placement of the FRB source in the host galaxy does not impact the mean of the host contribution across different redshifts [13]. Nonetheless, varying the feedback mechanism does impact the distrubtion of the host contribution. Therefore, Reischke, Kovač et al. (2024) provided an analytic model of the DM host contribution using the baryonificaiton correction model (A. Schneider, Kovač et al., 2025; A. Schneider and Teyssier, 2015). This model was able to reproduce the probability distribution of the host contribution given redshift and halo mass as found from the CAMELS simulation suit. However, integrating this impact in a cosmological analysis will double the dimensionality of the paramter space since another copy of baryonic feedback parameters are needed to account for the host contribution beside the cosmological contribution of the dispersion measure $\mathrm{DM}_{\mathrm{LSS}}$. In addition, the analytical model reuquires simplistic assumption regarding the distribution of stars in halos.

In this work, I assume that the host contribution follows a lognormal distribution since it fits the analytical prediction from Reischke, Kovač et al. (2024) and the simulation results from Theis et al. (2024) for a sample size of $10^4$ FRBs, which what we consider in this thesis. In general, if the lognormal distribution is parameterized by $\mu$ and $\sigma_{\mathrm{LN}}^2$, then the probability distribution funciton of the host contribution is given as

$$P_{\mathrm{host}}(\mathrm{DM}_{\mathrm{host}}; \mu, \sigma_{\mathrm{LN}}) = \frac{1}{\mathrm{DM}_{\mathrm{host}}\sigma_{\mathrm{LN}}\sqrt{2\pi}} \exp\left(-\frac{(\ln \mathrm{DM}_{\mathrm{host}} - \mu)^2}{2\sigma_{\mathrm{LN}}^2}\right), \tag{2.131}$$

where $\exp(\mu)$ and $\exp(2\mu + \sigma_{\mathrm{LN}}^2)\left[\exp(\sigma_{\mathrm{LN}}^2) - 1\right]$ are the median and the variance of the distribution.

- **Redshift uncertainty:** Follow up optical observations of FRBs allows spectroscopic estimation of the redshifts of its host galaxy. The measurement uncertainty on the redshift can induce additional scatter in the estimated cosmological covariance matrix and modify the source redshift distribution. The latter effect is especially important when cross-correlating FRBs with cosmic shear as performed by Reischke, Neumann et al. (2023). A faster way to localize FRBs is to use wide field spectroscopic or photometric surveys by deriving an astrometric solution, yielding a set of possible candidates of the host with a probability for each canditate (García et al., 2025). This method is, nonetheless, highly uncertain. Therefore, the redshift uncertainty is an important systematic uncertainty to mitigate in an analysis involving FRBs.

  In this work, I assume that the measured that a redshift measuremnet $z_i$ is sampled from a Gaussian distribution with mean $z_i$ and standard deviation $\sigma_i = \eta z_i$, where $\eta_i$ is a number between 0 and 1.

- **Selection effects:** The time resolution of radio telescope and its ability to detect a wide range of

[13] Sources that lie at the further end of the galaxy along the line of sight will pick up higher contribution as compared to a source at the near end of a galaxy, however, on average, the mean of the distribution does not change.

DM values is limited by the instrument. In other words, radio telescopes are only able to detect DM values up to a certain upper limit. THis effect originate from dispersive smearing in the different radio frequency channels in the telescop, leading to an artificial broadening of the signal and low signal-to-noise ratio (Hoffmann et al., 2025). In general, this affects the normalization of the PDF of measured DM and not the shape. Therfore, the search efforts are restricted to a $\mathrm{DM}_{\mathrm{max}}$, depending on the instrument. Table 2.1 shows some exmaples for radio surrveys and their correponding $\mathrm{DM}_{\mathrm{max}}$. Since DM is an integrated value, a lower value of $\mathrm{DM}_{\mathrm{max}}$ restricts the search to inlcude low redshifts only.

| Survey | $\mathrm{DM}_{\mathrm{max}}$ (pc cm$^{-3}$) |
|---|---|
| DSA-110 | 1500 |
| FAST/CRAFTS | 5000 |
| FAST/GPPS | 3700 |

Table 2.1: Maximum dispersion measure search limits for FRB surveys, the values are adopted from Hoffmann et al. (2025)

The impact of selection was studied by Sharma, Ravi et al. (2026). In this work, the authors use a probabilistic model to describe the DM with all the possible selection effect. Beside the selection based on $\mathrm{DM}_{\mathrm{max}}$, the authors consider the redshift source distribution by modeling with 3 free parameters and fitting this model to data along with the cosmological and baryonic feedback parameters. Furthermore, this work concludes that selection effects lead to $3\sigma$ bias in the inferred paramters for $10^4$ FRBs.

In this thesis, I fix the redshift source distribution to coincide with that expected from the Square Kilometer Array Observatory (SKAO) (Braun et al., 2015), and impose a smooth cut on the measured $DM$ by multiplying the PDF $P(\mathrm{DM}^i; \{z^i_{\mathrm{FRB}}\})$ with a sigmoid kernel (Shannon et al., 2018) parameterized by $\mathrm{DM}_{\mathrm{max}}$ and the smoothness of the cut $\epsilon$

$$\sigma(\mathrm{DM}; \mathrm{DM}_{\mathrm{max}}, \epsilon) = \frac{1}{1 + \exp\left(\frac{\mathrm{DM}-\mathrm{DM}_{\mathrm{max}}}{\epsilon}\right)} \tag{2.132}$$

Therefore, for a simulated catalog, each value of $DM$, is weighed with $\sigma(\mathrm{DM}; \mathrm{DM}_{\mathrm{max}}, \epsilon)$.

- **Unlocalised hosts:** So far, the number of detected FRBs is 3000[14], and only 100 of them has identified host (Reischke and Hagstotz, 2025) i.e only the redshifts of 100 FRBs is known. This poses as an obstacle to use all FRBs for cosmological inference since redshifts are important to calculate data covariance and constrain the DM-$z$ relationship. Since follow-up optical observation to identify the host galacxy is both time and resource consuming, it is important to realised the best strategy to distribute these resources. In this thesis, I explore whether constraining the DM-$z$ relation is possible if only a subset of the observed FRBs has an identified host. I explore the following cases
  - A random subset of FRBs has no identified hosts.
  - A subset of high redshfit FRBs has no identified hosts.

[14] CHIME/FRB Catalog 2.

- A subset of low redshfit FRBs has no identified hosts.

Retrospectively, FRBs with high redshift has higher signal-to-noise since they pick higher contribution from the electorn density along their path to Earth, however, the measurement uncertainty of their redshift is high. Meanwhile, FRBs with low redshifts has lower signal-noise-ratio since the path they travel is less as compared to high redshift sources, however, their measured redshfits have lower uncertainties. The results will show whether observing high- or low- redshift sources will be the best stretegy to follow for follow-up optical observations.

CHAPTER 3

# Methodology

In this section, I explain the methoddology used in this thesis. The end purpose is to show how the $DM$-$z$ relation deviates from its what is expected by contaminating mock FRBs catalog with systematic uncertainties through simulating FRBs catalogs and studying the impact of systematic uncertainties when contaminating catalogs with these systematic uncertainties. Fig. 3.1 shows a flowchart of how to generate a mock catalog. I explain the steps in details below. I perform the analysis assuming the detection for $N_{\mathrm{FRBs}} = 10^4$. This is the expected number of detected FRBs per year by future FRBs survey such as ther SKAO (Macquart et al., 2015). Future work should focus on scaling up to larger sample size since the impact of systematic uncertainties changes with sample size.

Table 3.1 shows the cosmological and baryonic feedback paramters which are used to generate mock catalogs. These parameters are used to obtain an expectation of the electron power spectrum.

To model the dispersion measure signal, I use the Generator for Large-Scale Structure (GLASS) (Tessore et al., 2023) package. I describe this package in further details in section 3.2.1. Briefly, the GLASS package generates lognormal maps of a cosmological field starting from a given angular power spectra. On contrary to the commonly used `healpy` package, it assumes that the cosmological field follows a lognormal distribution. Though this is not the case in for a real cosmological field, the GLASS package preserves the two-point statistics of the cosmologocal field, which is enough for the sample size I am considering in this thesis. Higher order statistics such as the bispectrum (3-point) and the trispectrum (4-point) requires a larger sample size for detection. Using the electron power spectrum, I project the spectra on the unit sphere to obtain an expectation of the angular power specta in bins of redshifts. After that, I feed the resultatnt power spectra to GLASS in order to obtain maps of electron density in bins of redshift. THe next step constitute in sampling randomly on the sky and according to the source redshift distribution in redshift then sum the contribution from different shells. In general, the simulation spicificaitons are shown in Table 2.3(b)

## 3.1 Modelling the electron power spectrum

To model the electron power spectrum, I use the halo model (introduceed in section 2.6), where I obtain halo profiles using the Baryonification Model (BCM) (A. Schneider and Teyssier, 2015; A. Schneider, Kovač et al., 2025).

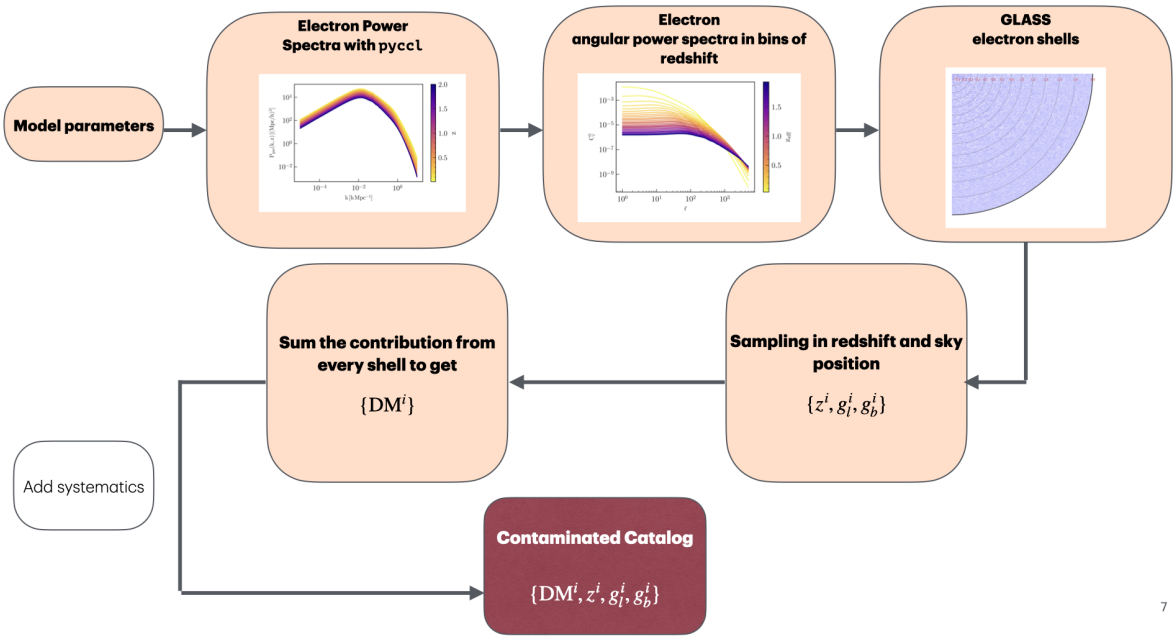



Figure 3.1: A flowchat of the simulation pipeline introduced in this section. Starting from model parameters I obatin the non-linear electron power spectrum via the halo model. Then, I perform the Limber integral to obatin the electron angular power spectrum in each redshift bin. After that, I use the GLASS package to obatin lognormal maps of electrons in bins of redshifts. Then, I sample in angular positions and redshifts and sum the contribution of each shell to obtain an uncontaminated catalog. Eventually, The systematic uncertainties are then added.

| Parameter | Description | Fixed value | Prior |
|---|---|---|---|
| $\Omega_b$ | Baryon density parameter | 0.0493 | - |
| $\Omega_c$ | Cold dark matter density parameter | 0.264 | - |
| $h$ | Dimensionless Hubble parameter | 0.6766 | - |
| $n_s$ | Spectral index of primordial power spectrum | 0.9665 | - |
| $\sigma_8$ | Amplitude of matter fluctuations | 0.8102 | - |

Table 3.1: A table with the cosmological parameters used in this thesis. The values are adopted from Planck 2018 results (Aghanim et al., 2020).

### 3.1.1 The baryonification model

This section is based on the review of the baryonificaiton correction model presented in Torkamani et al. (2026).

The purpose of the baryonificaiton model (BFC) is modify the profiles of dark matter halos from N-body gravitational simulation to that which involves baryonic feedback. The change is expressed as a shift in the position of dark matter particles which is parameterized by a set of baryonic feedback parameters.The displacement for dark matter (dm) and baryonic matter (bar) is performed along the radius of a dark matter halo. In practice, the dm and the bar profiles are modified differently when baryonic feedback is taken into consideration. In this thesis, I only focus on the case of modified bar

| Property | Description | Value |
|---|---|---|
| $z_{\rm min}$ | Minimum redshift | 0.01 |
| $z_{\rm max}$ | Maximum redshift | 2.0 |
| $n_{\rm shells}$ | Number of shells | 30 |
| $\ell_{\rm max}$ | Maximum multipole | 4100 |
| $N_{\rm side}$ | Map resolution (HEALPix) | 4096 |
| $b_{\rm h}(M)$ | Bias model as a funciton of halo mass $M$ | J. L. Tinker et al. (2010) |
| $n(M)$ | Halo mass function | J. Tinker et al. (2008) |

Table 3.2: The specifiactions of FRB catalog simulations. HEALPix (Gorski et al., 2005) is an algorithm to pixellize as an image with certain resolution. The resolution parameter $N_{\rm side}$ can be understood as the inverse of the diameter of one pixel on the map $\theta_{\rm min} \approx 1/N_{\rm side}$.

profile. In a a scenario without baryonic feedback, the baryons follow the distribution of dark matter. The baryon halo profile $\rho_{{\rm bar},i}$ can be written in terms of the underlying dark matter profile $\rho_{\rm NFW}$[1] on small scales in addition to a large scale contribution from the matter outside of the collapsed halo region which I label as $\rho_{\rm 2h}$[2]. In total, the final expression reads

$$\rho_{\rm bar,i}(r) = f_{\rm bar}\left[\rho_{\rm NFW}(r) + \rho_{\rm 2h}(r)\right], \tag{3.1}$$

with $f_{\rm bar} = \Omega_{\rm b}/\Omega_{\rm c}\ \rho_{\rm 2h}$ given in term the linear order 2-point correlation function $\xi_{\rm lin}$

$$\rho_{\rm 2h}(r) = \left[1 + b(\nu)\xi_{\rm lin}(r)\right]\Omega_{\rm b}\rho_{\rm crit}, \tag{3.2}$$

with $b(\nu)$ the halo bias taken from J. L. Tinker et al. (2010). $\nu$ in this context refer to the ratio between the critical density contrast for spherical collapse at linear order $\delta_{\rm c,lin}$ and the standard deviation of halo mass $\sigma(M)$.

Since baryonic feedback affects the distribution of matter on small sacles, the modified baryon profile the component $\rho_{\rm NFW}$[3] should be modified. Empirically, this mean that on small scales, the contribution to the baryon profile comes primarily from gas and stars within a dark matter halo. The modified component $\rho_{\rm bar,f}$ becomes

$$\rho_{\rm bar,f}(r) = \rho_{\rm gas}(r) + \rho_{\rm star}(r) + f_{\rm bar}\rho_{\rm 2h}(r). \tag{3.3}$$

Each of the densities in the previous equation is parametrized with a set of cosmological and baryonic feedback parameters. For any species u, the cumilitive mass up to a radius $r$ is given By

$$M_{\rm u}(R) = 4\pi\int_0^R r^2\rho_{\rm u}\,{\rm d}r = f_{\rm u}Y_{\rm u}(R), \tag{3.4}$$

where $f_{\rm u}$ is the abundance of a component (relative to the total matter) and $Y_{\rm u}$ is normalized such that

[1] $\rho_{\rm NFW}$ is an analytical expression of dark matter halo density profiles, predicted empirically by fitting the expression to N-body gravitional simulation (Navarro, Frenk and White, 1997)

[2] This contribution is labelled similary to the 2-halo term of the power spectrum since it dominates the contribution on large scales similar to the 2-halo term of the power spectrum

[3] $\rho_{NFW}$

$Y_{\rm u}(\inf) = M_{\rm tot}$ To qunatify the deviation $d(r)$ in a particle's position between its initial (gavitational effects only) position $r_{\rm bar,i}$ to its final (gravitional effects and baryonic feedback) position $r_{\rm bar,f}$, the radius $r_{\rm bar}$ as a function of the cumilitive mass $M_{\rm bar}$ is obtained

$$r_{\rm bar} = {\rm inv}\left[M_{\rm bar}(r)\right], \tag{3.5}$$

and the resultant deviation is written as

$$d(r) = r_{\rm bar,f} - r_{\rm bar,i}. \tag{3.6}$$

The mapping from $r_{\rm bar,i}$ to $r_{\rm bar,f}$ is parameterized with the baryonic feedback paramters shown in Table 3.3. In addition, Fig. 3.2 shows how the BFC model changes the profiles of dark matter halos.

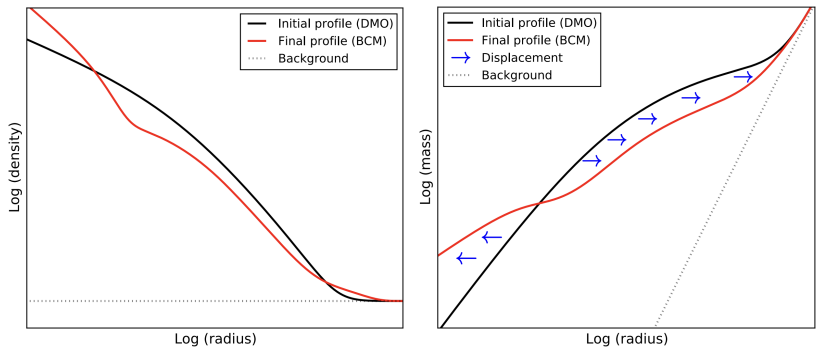



Figure 3.2: A plot showing how the BFC model modifies halo profiles (left) and the cumilitive mass (right). At small radius, the profile shifts up to account for star formation processes while it shiffts down on larger radius to account for AGN and supernova feedback. The figure is adopted from A. Schneider and Teyssier (2015).

In this thesis, I use the halo gas profiles predicted by the BFC model to obatin a theoretical expectation of the gas power spectrum, which in turn is the electron power spectrum. I derive the 1-halo and the 2-halo terms using the Core Cosmology Library which is discussed in the next section. As an example of how baryonic feedbck modifies the power spcetrum, Fig. 2.3(b) shows how the power spectrum changes with changing values of $\log_{10} M_c$, one of the baryonic feedback parameters.

### 3.1.2 The Core Cosmology Library (CCL)

The Core Cosmology Library (CCL) (Chisari et al., 2019) is a package that enables fast and efficient calculations of basic cosmological observables with high accuracy. CCL implements a Boltzmann solver in order to obtain a prediction of the linear power spectrum on a grid of redshifts $z$ and scale $k$. It also provides a bicubic interpolator for accurate calculation of the power spectrum, starting from the initial grid points. In this thesis, I sample 30 grid points in $z$ and 150 points in $k$. These are enough grid points to ensure sub-percent accuracy as compared larger numbe of grid points. However, since the $\log P(k, z)$ is linearly related to $\log k$ for both large and small values of $k$, it is possible to extrapolate the power spectrum in logarithmic space beyond the ranges specified in Table 3.2. I show the extrapolation in Fig. **??**. In addition, to obtain the non-linear power spectrum, CCL implements the halo model where the 1-halo and the 2-halo terms are calculated using the halo profiles obtained from the BFC model. The

| Parameter | Prior | Description | Fixed value |
|---|---|---|---|
| $\theta_{\rm co}$ | U[0.001, 0.5] | Core radius of the gas profile relative to $r_{200}$ (Eq. 2.6) | 0.3 |
| $\log_{10} M_{\rm c}$ | U[11.0, 15.0] | Pivot mass for the mass dependent slope of the gas profile (Eq. 2.6 & Eq. 2.7) | - |
| $\mu$ | U[0.0, 2.0] | Transition steepness of mass dependent gas parameter (Eq. 2.6 & Eq. 2.7) | - |
| $\delta$ | U[1.0, 11.0] | Characterizes the transition from inner to outer part of the gas profile (Eq. 2.6) | - |
| $\eta$ | U[0.0, 0.5] | Total stellar fraction within a halo (Eq. 2.10) | 0.1 |
| $d\eta$ | U[0.0, 0.5] | Fraction of central galaxies w.r.t. total stellar fraction (Eq. 2.10) | 0.22 |
| $N_{\rm star}$ | U[0.0, 0.05] | Stellar fraction amplitude (Eq. 2.10) | 0.030 |
| $c_{\rm iga}$ | U[0.0, 1.0] | Cold gas fraction (see Eq. 2.12) | 0.1 |
| $b_{\rm hse}$ | $\mathcal{N}(0.26, 0.07)$ | Hydrostatic mass bias | - |
| $\alpha_{0,\rm nt}$ | - | Non-thermal pressure amplitude (see Eq. 2.15) | 0.1 |

Table 3.3: The parameters of the BFC model.

halo mass function and the halo bias are taken from J. Tinker et al. (2008) and J. L. Tinker et al. (2010), respectively. I consider only a linear bias model where the term $\beta$ from Eq. 2.62 is set to 0. I show a plot of th epower spectrum for different redshfits in Fig. **??**.

In this thesis, I use CCL to obtain a non-linear prediction of the elctron power spectrum using the halo model where the integrals in Eqs. 2.70 and 2.71 are calculated using the halo profiles obtained from the BFC model. I use the resultant power spectrum to obtain an expectation of the electron angular power spectrum in bins of redshifts using the Limber approximation, which is described in the next section.

### 3.1.3 The electron angular power spectrum

To obtain the electron angular power spectra in bins of redshifts, I define the electron density contrast $\delta_{\rm e}^i$ in a redshift bin $i$ as

$$\delta_{\rm e}^i(\hat{\mathbf{x}}) = \int W^i(z)\, \delta_{\rm e}(z, \hat{\mathbf{x}})\, {\rm d}z, \tag{3.7}$$

with $W^i(z)$ is the normalized weight function of the redshift bin $i$. The general form of the weight funciton is a top-hat function multiplied by a weight factor

$$W_i(z) = \begin{cases} f(z) & \text{if } z_{i-1} \leq z < z_i, \\ 0 & \text{otherwise,} \end{cases}, \tag{3.8}$$

where the form of the function $f(z)$ is chosen arbitrarly such as to account for the expansion of the Universe. I discuss the choice of $f$ in section 3.2.1. The electron angular power spectrum $C_{\delta_e \delta_e}^{(ij)}(\ell)$

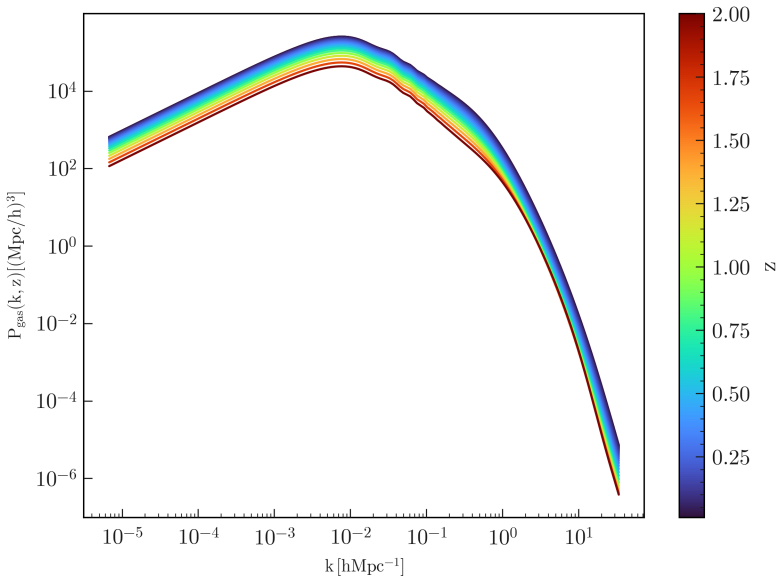



Figure 3.3: A plot showing the power spectrum in logarithmic space. The power spectrum is extrapolated in logarithmic space beyond the range of $k$ and $z$ specified in Table 3.2.

between two redshift bins $i$ and $j$ is given by

$$C^{(ij)}_{\delta_e\delta_e}(\ell) = \frac{2}{\pi}\int \mathrm{d}\chi \frac{\mathrm{d}z}{\mathrm{d}\chi}\frac{\mathrm{d}z}{\mathrm{d}\chi} W^{(i)}(z(\chi)) \int \mathrm{d}\chi' \frac{\mathrm{d}z}{\mathrm{d}\chi'} W^{(j)}(z(\chi'))$$
$$\int \mathrm{d}k\, k^2\, P_{\mathrm{e,nl}}(k, z(\chi), z(\chi'))\, j_\ell(k\chi)\, j_\ell(k\chi'), \tag{3.9}$$

where this derivation follows similarly to the derivation of the covariance matrix in section 2.9. Using the Limber approximation, the expression is simplified into

$$C^{(ij)}_{\delta_e\delta_e}(\ell) = \int \mathrm{d}\chi \frac{\mathrm{d}z}{\mathrm{d}\chi}\frac{\mathrm{d}z}{\mathrm{d}\chi}\frac{W^{(i)}(z(\chi))W^{(j)}(z(\chi))}{\chi^2} P_{\mathrm{e,nl}}\left(k = \frac{\ell + 1/2}{\chi}, z(\chi)\right). \tag{3.10}$$

## 3.2 Simulating the dispersion measure signal

Add a table with the fiducial values and their prior. explain how we ignore the redshift evolution of some parameters. Motivate the choice for the number of shells and the redshift range.

### 3.2.1 The Generator for Large Scale Structure (GLASS)

discuss the choice of the the weight function mention the reason behind the choice of 30 shells. and the binning in comoving space. add a plot of the concentric shells. speak a bit about HEALPix add a plot of the shells.

### 3.2.2 The numerical implementation of the cosmological covariance matrix

- Add a plot for the validity of the divisions when implementing in z.

## 3.3 Simulating the systematic uncertainties

Table 3.4: Summary of Model Parameters

| Symbol | Description | Range |
|---|---|---|
| | *Redshift Uncertainty* | |
| $\eta$ | the ratio between a redshift measurement $z$ and its standard deviation | $[0.05, 0.4]$ |
| | *The host contribution* | |
| $\mathrm{DM}_{host,mean}$ | The underlying Gaussian mean of the lognormal distribution | $[50, 300]$ |
| $\sigma_{\mathrm{LN}}$ | The scale of the lognormal distribution | [0.01, 0.15] |
| | *The Milkyway contribution* | |
| $\mathrm{DM}_{\mathrm{MW}}(\hat{\mathbf{x}})$ | The Milkyway contribution as a function of angular position $\hat{\mathbf{x}}$ | Given by the NPE2001 model |
| | *Selection effects* | |
| $\mathrm{DM}_{\mathrm{max}}$ | The maximum value of DM that the instrument is able to detect. | [1500, 3500] |
| $\epsilon$ | The smoothing scale of the PDF. | [0.1, 3] |
| | *Missing redshifts* | |
| $f$ | The fraction of FRBs with missing redshift. | [0.05, 0.4] |

Table 3.5: A table with the parameters of the systematic uncertainties. The range of the parameters is chosen to be wide enough to cover the expected values for these parameters.

### 3.3.1 Milkyway contribution

### 3.3.2 Host contribution

offer an explnation for the chosen values of the parameters.

### 3.3.3 Redshift uncertainty

I simulate the imapct of redshift uncertainty by assuming that a redshift measuremnet $z_i$ is sampled from a Gaussian distribution with mean $z_i$ and standard deviation $\sigma_i = \eta z_i$, where $\eta_i$ is a number between 0 and 1. To propagate the error on the cosmological covariance, consider that $(z_i, z_j) \sim \mathcal{N}((\hat{z}_i, \hat{z}_j), \Sigma)$, with $\mathcal{N}$ is the normal distribution and

$$\Sigma = \begin{bmatrix} \sigma_i^2 & 0 \\ 0 & \sigma_j^2 \end{bmatrix} \tag{3.11}$$

To propagate the error, I calcualte the expectation value $\langle \hat{C}_{ij} \rangle$, given as

$$\langle \hat{C}_{ij} \rangle = \int \mathcal{N}((\hat{z}_i, \hat{z}_j), \Sigma)\ C_{ij}\ dz_i\ dz_j \tag{3.12}$$

With Taylor's expansion around $(\hat{z}_i, \hat{z}_j)$ and setting $\Delta_{ij} = (z_i, z_j) - (\hat{z}_i, \hat{z}_j)$, the covariance matrix can be written

$$\langle \hat{C}_{ij} \rangle = \int \mathcal{N}\ \left( \hat{C}_{ij} + \nabla C_{ij}|_{(\hat{z}_i,\hat{z}_j)} \Delta_{ij} + \frac{1}{2} \Delta_{ij}^T \nabla^2 C_{ij}|_{(\hat{z}_i,\hat{z}_j)} \Delta_{ij} + ..... \right)\ dz_i\ dz_j. \tag{3.13}$$

Since the Gaussian distribution has vanaishing moments for moment larger than third order, the expression is simplified into

$$\langle \hat{C}_{ij} \rangle = \hat{C}_{ij} + \frac{1}{2}\ \sigma_i^2\ \partial_{ii}^2\, C_{ij}|_{(\hat{z}_i,\hat{z}_j)}\ + \frac{1}{2}\ \sigma_j^2\ \partial_{jj}^2\, C_{ij}|_{(\hat{z}_i,\hat{z}_j)}. \tag{3.14}$$

Therefore, it is possible to account for the error in the covariance due to redshift uncertainty using the second derivative of the covariance with respect to redshift $z$.

I show the impact of redshift uncertainty on the scatter in the DM-$z$ relation in chapter 4.

### 3.3.4 Selection effects

- show a plot of full PDF and a truncated PDF. - Discuss the choice for different normalizations and the excluded FRBs depending on redshift.

### 3.3.5 Unlocalised hosts

I simulate unlocalised FRBs in a fashion similar to what is descibed in seciotn 2.10. To fill in missing redshift, I use the fiducial parameters in order to obtain an expectation of the unknown redshift $z_i = z(\mathrm{DM}_i)$ from the measured $\mathrm{DM}_i$. To propagate the error to the redshift space, I assume that DM follows a lognormal distribution with underlying Gaussian mean $\mathrm{DM}_i$ and variance $\sigma^2(z_i)$, such that

$$\mathrm{DM}_i \sim P(\mathrm{DM}; \mu^i, \sigma^2_{\mathrm{LN},i}) \tag{3.15}$$

with

$$\mu^i = \ln\left( \frac{\mathrm{DM}_i^2}{\sqrt{\sigma^2(z^i) + \mathrm{DM}_i^2}} \right),\ \sigma^2_{\mathrm{LN},i} = \ln\left( 1 + \frac{\sigma^2(z^i)}{\mathrm{DM}_i^2} \right) \tag{3.16}$$

The lognormal assumption is valid in the case of this work since the DM contributions from the different shells are lognormal. After that, I sample sample from the distribution, and I use the same mean DM-$z$ relation to map the sampled DM to $z$-space. After that, I calculate the sample variance of the mapped redshifts $\sigma_i^2$, where $\sigma_i/\sqrt{n_{\mathrm{sample}}}$ is the error on the estimated redshift $z_i$.

To propagate the error to the covarirance, I use the same formalism introduced in section 3.3.3, only as an approximation, since the underlying PDF of the estimated redshift is not a Gaussian distribution. For the values of $\sigma_{ii}$, if the FRB has an undefined host, the redshift uncertainty is calculated with the method introduced before. Meanwhile, if the redshift is exact, then I set the value of $\sigma_{ii} = 0$. In this theis, I aim to disentangle the different systematic uncertainties. In reality, These effects could be degenerate in the

way they impact the measured DM or the covariance. I discuss that futher in chapter 5.

## 3.4 Data Compression

I simulate the electron density shells and place FRBs in the same redshift and angular position as in the mock catalog. This will reduce the computational cost and the need to calculate the covariance matrix for each realisation within the SBI pipeline. (Try to do this excercise without choosing the sdame redshifts for a lower number of sources such that it lowers down the computational costs)

#### Numerical derivative stability

#### The Sensitivity of compressed statistics to model parameters

Add a plot about derivative stability. Add a plot for the sensitivity.

## 3.5 Simulation-based inference

### 3.5.1 Approximate Baysian inference

### 3.5.2 Neural Posterior Estimator

### 3.5.3 The Sequentual Neural Posterio Estimator

#### The truncated sequential neural posterior estimator

### 3.5.4 Flow matching posterior estimator

CHAPTER 4

# Results

Add a plot of Chi-squared vs. the parameters of the systematic uncertainties.

CHAPTER 5

# Discussion & Conclusion

talk about how different systematic uncertainties could be degenerate.

APPENDIX A

# Useful information

- Fourier transform conventions. - Plane wave decomposition into spherical harmonics in the derivaiton o the cosmological covariance. Look at equation 13 on these lecture notes https://scipp-legacy.pbsci.ucsc.edu/ haber/ph215/PlaneWaveExpansion.pdf. - Definition of the Legendre polynomial. - Virial theorem. - spherical collapse model and the critical density. - The Dirac delta function. - The limber approximation. - The 4 velocity of massless particles (necessairy for the the derivtion fo the dispersion measure expression)

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